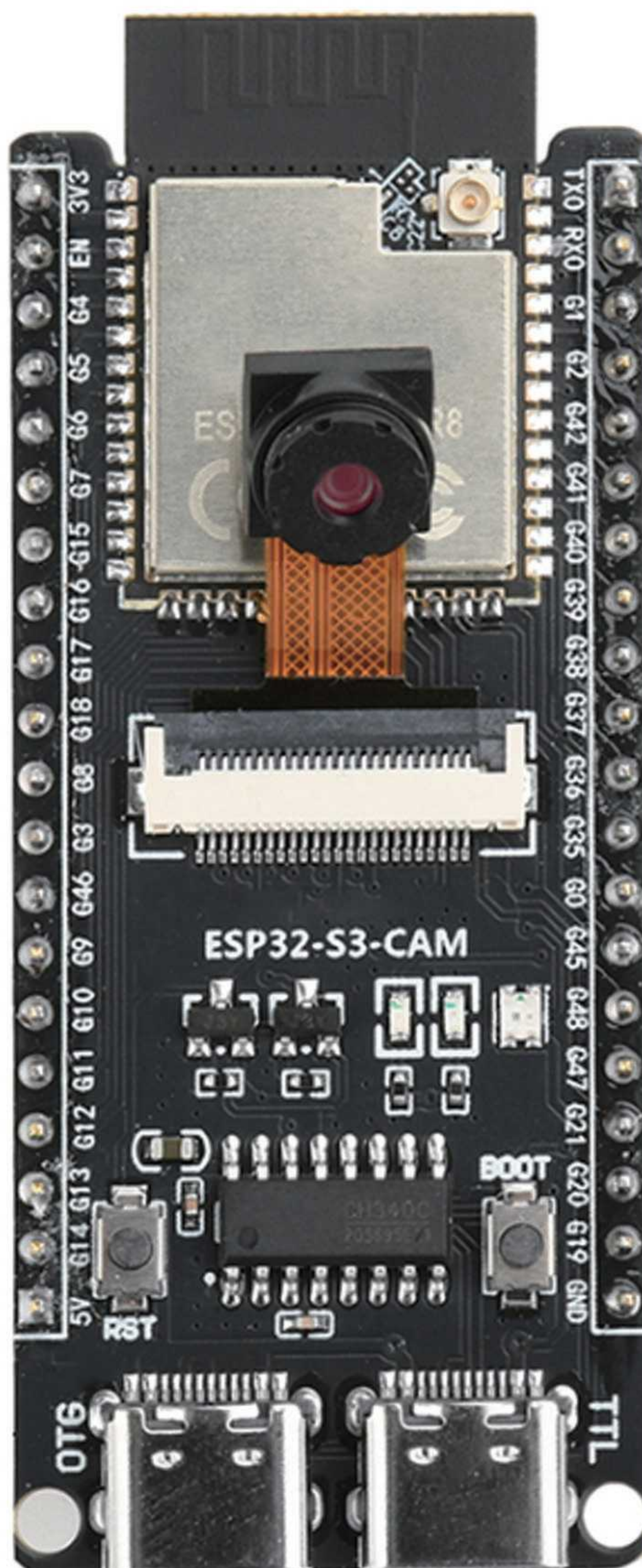


ESP32 S3 CAM Development Board



Catalogue

Product Introduction	1
Pinouts	2
Dimension	3
instructions	4

Product Introduction

ESP32-S3 integrates 2.4GHz Wi Fi (802.11b/g/n) and supports 40MHz bandwidth; Its low-power Bluetooth subsystem supports Bluetooth 5 (LE) and Bluetooth Mesh, and can achieve long-distance communication through Coded PHY and broadcast extension. It also supports 2Mbps PHY to improve transmission speed and data throughput. The ESP32-S3 has superior Wi Fi and Bluetooth LE RF performance and can operate stably even at high temperatures.

Product features:

1. Excellent Wi Fi performance: Integrated with IEEE 802.11b/g/n protocol to ensure high-speed and stable connection.
2. Advanced Bluetooth subsystem: Supports Bluetooth 5 (LE) and Bluetooth Mesh for long-distance communication.
3. Efficient data processing capability: Built in 32-bit LX7 dual core processor, providing powerful computing performance and a balance of low power consumption.
4. Integration of multiple wireless functions: Wi Fi and Bluetooth LE share the same antenna, optimizing multitasking.
5. High temperature stable operation: With superior performance, it can maintain stable operation even in high temperature environments and is suitable for various harsh working environments.

Performance parameters:

Supports IEEE802.11b/g/n protocol

Supports 20MHz and 40MHz bandwidth in the 2.4GHz frequency band

Supports 1T1R mode with a data rate of up to 150Mbps

Wireless Multimedia (WMM)

Frame Aggregation (TX/RX A-MPDU, TX/RX A-MSDU)

Immediate BlockACK

Fragmentation and reassembly

Low power Bluetooth (BluetoothLE): Bluetooth5, BluetoothMesh

High power mode (20dBm)

Supports speeds of 125Kbps, 500Kbps, 1Mbps, 2Mbps

Wi Fi and Bluetooth coexist, sharing the same antenna

32-bit LX7 dual core processor, with a maximum clock speed of 240MHz

WIFI:

Supports IEEE 802.11b/g/n protocol

Supports 20 MHz and 40 MHz bandwidth in the 2.4 GHz band

Supports 1 T1R mode with a data rate of up to 150 Mbps

Wireless Multimedia (WMM)

Frame Aggregation (TX/RX A-MPDU, TX/RX A-MSDU)

Immediate Block ACK

Fragmentation and reassembly

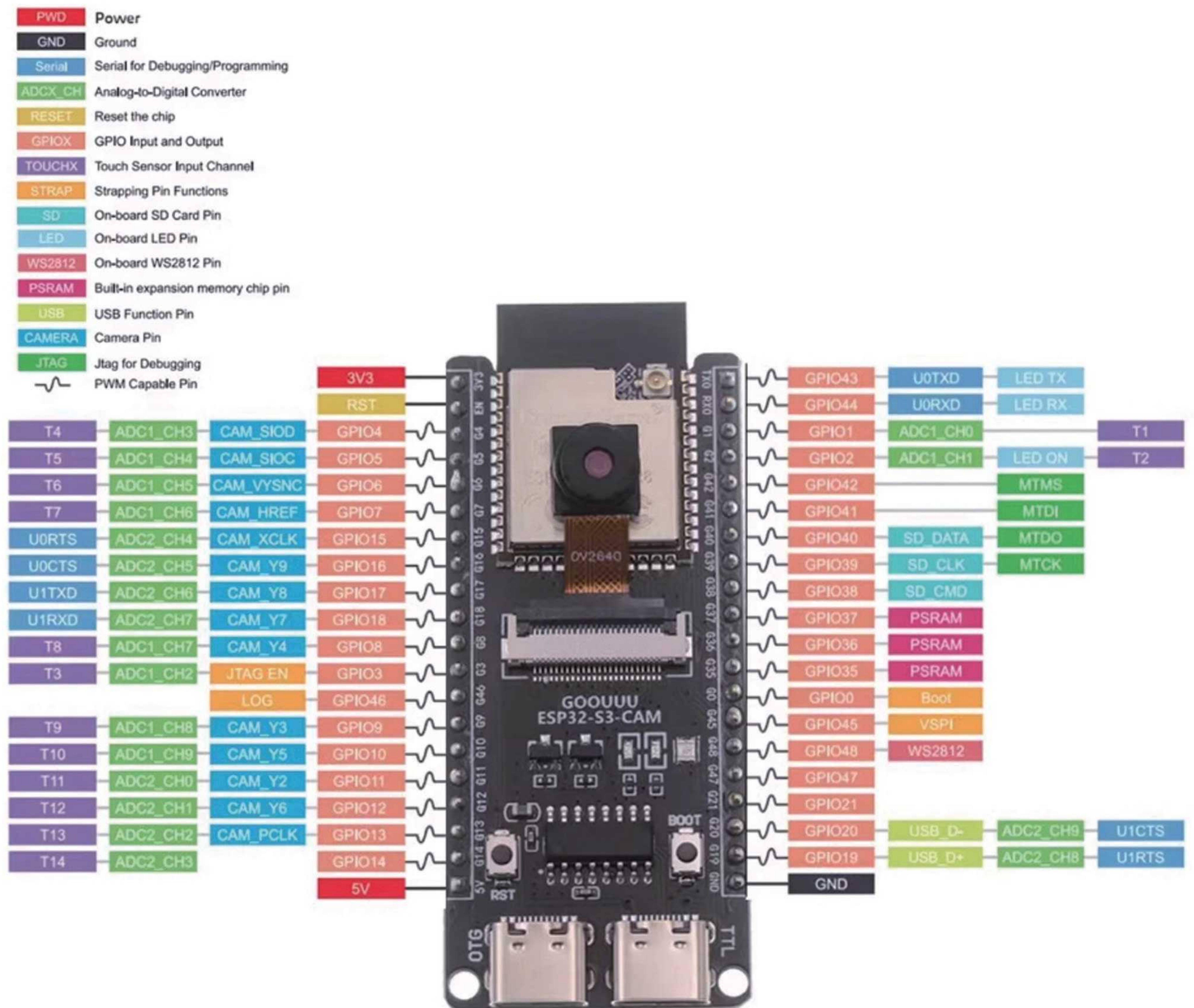
Beacon automatic monitoring (hardware TSF)

4 virtual Wi Fi interfaces

Simultaneously supporting Infrastructure BSS Station mode Please note the SoftAP mode and Station+SoftAP mode, When ESP32-S3 scans in Station mode, The SoftAP channel will change simultaneously

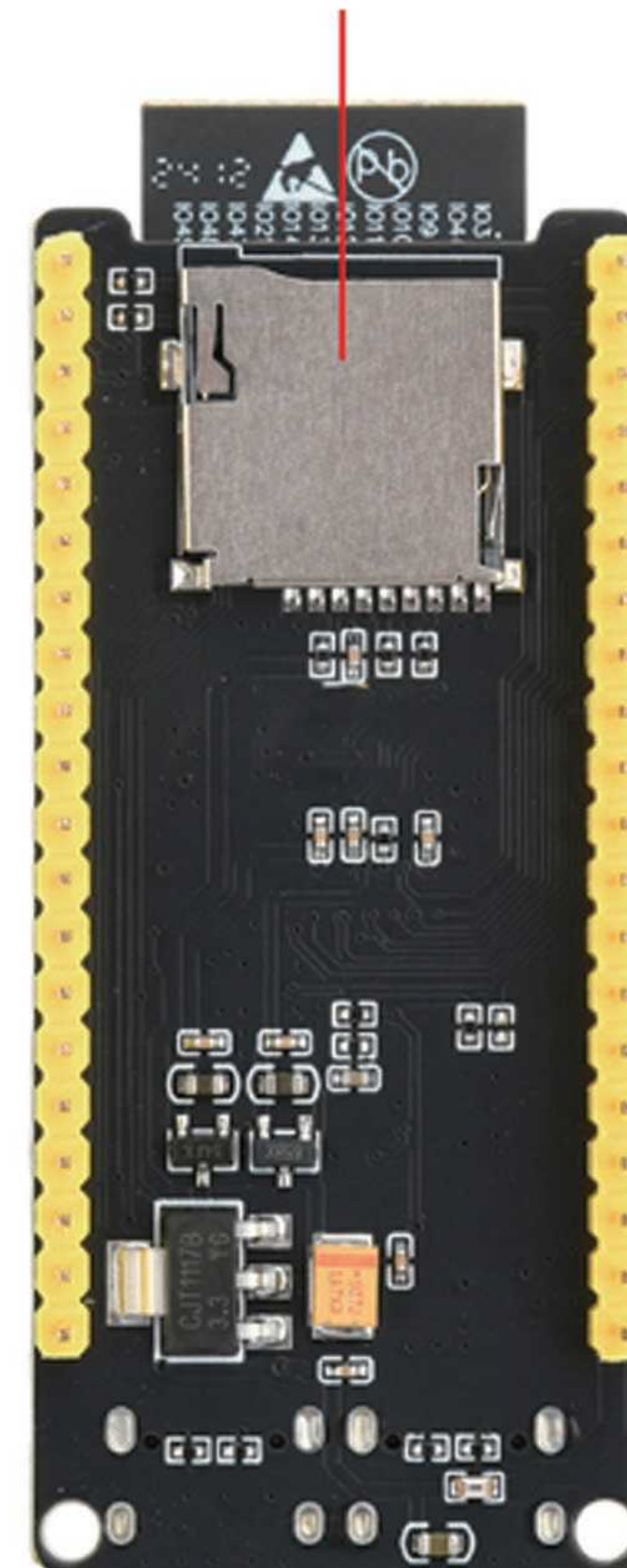
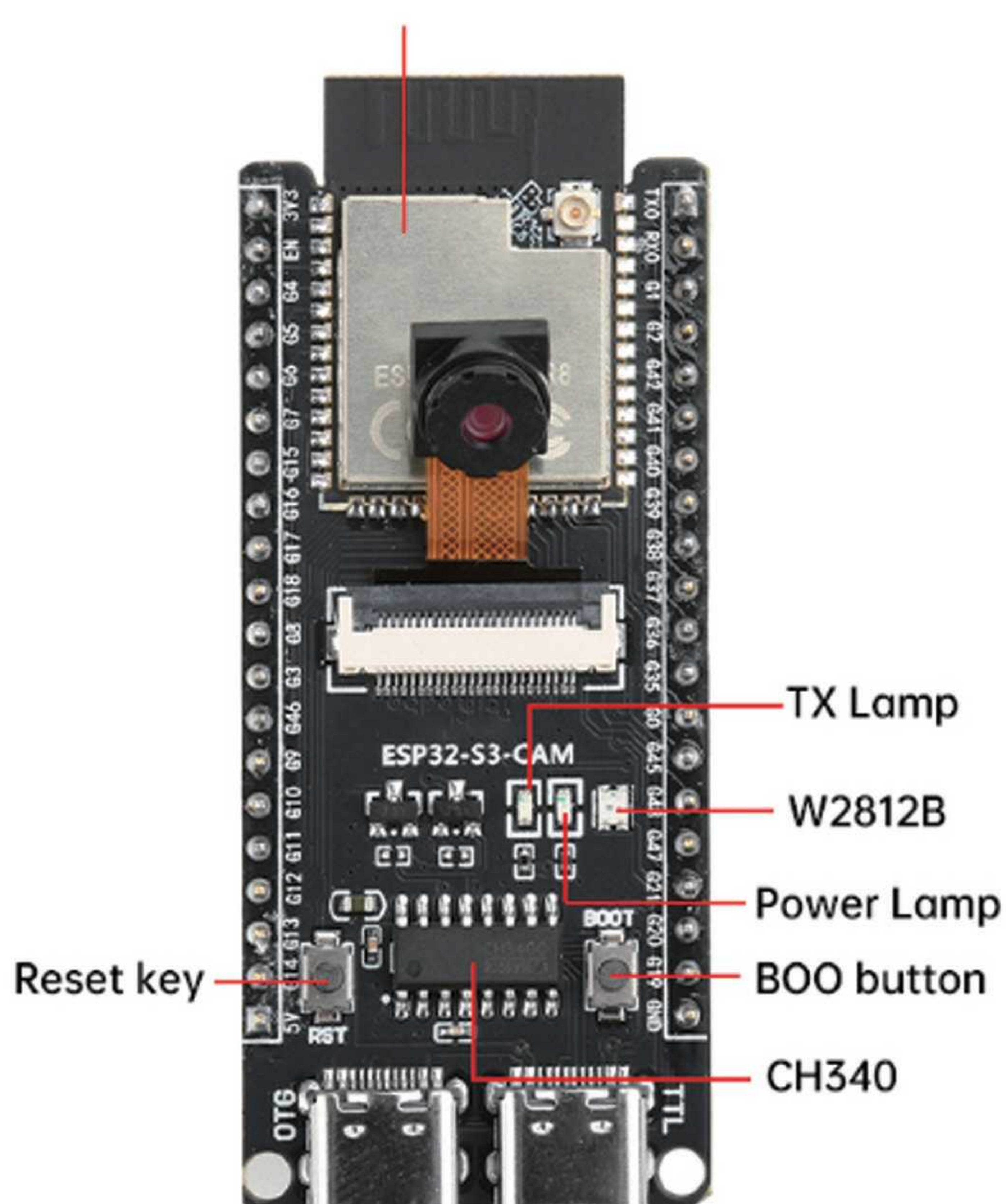
PCB board size: 62.6mm * 28.3mm

Product Pin Description

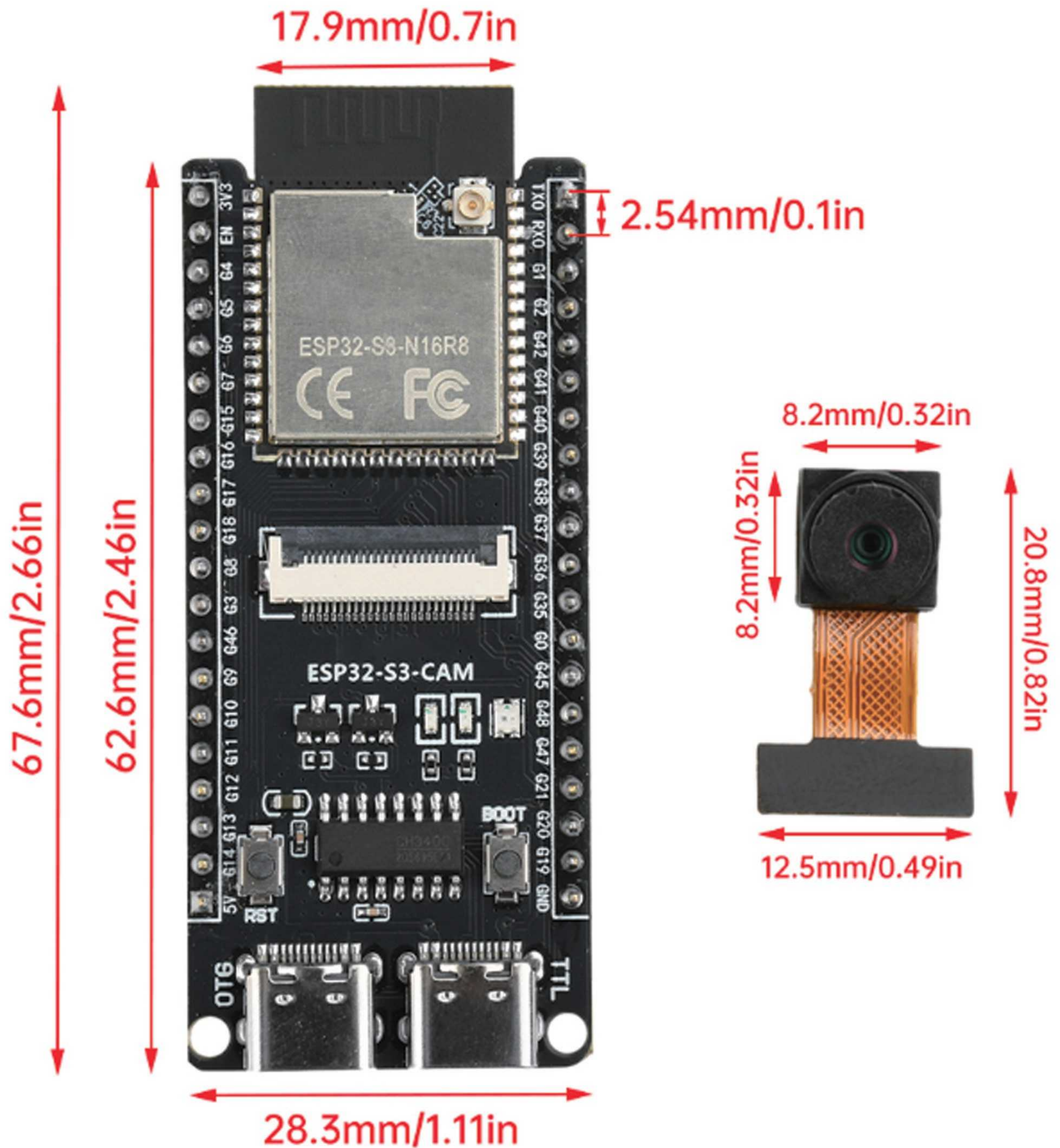


ESP32-S3 N16R8

TF card holder (back)



Product size

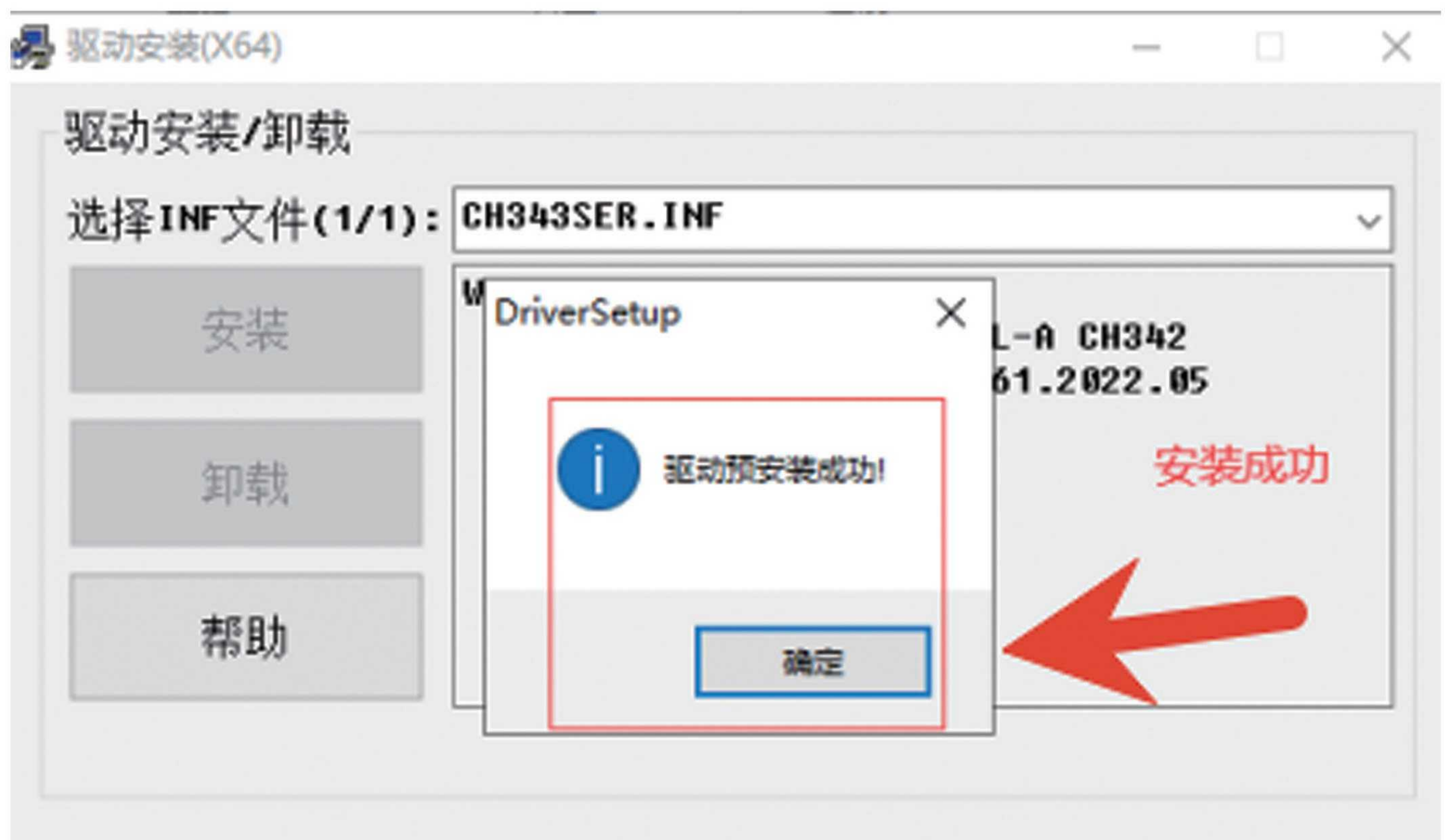


instructions

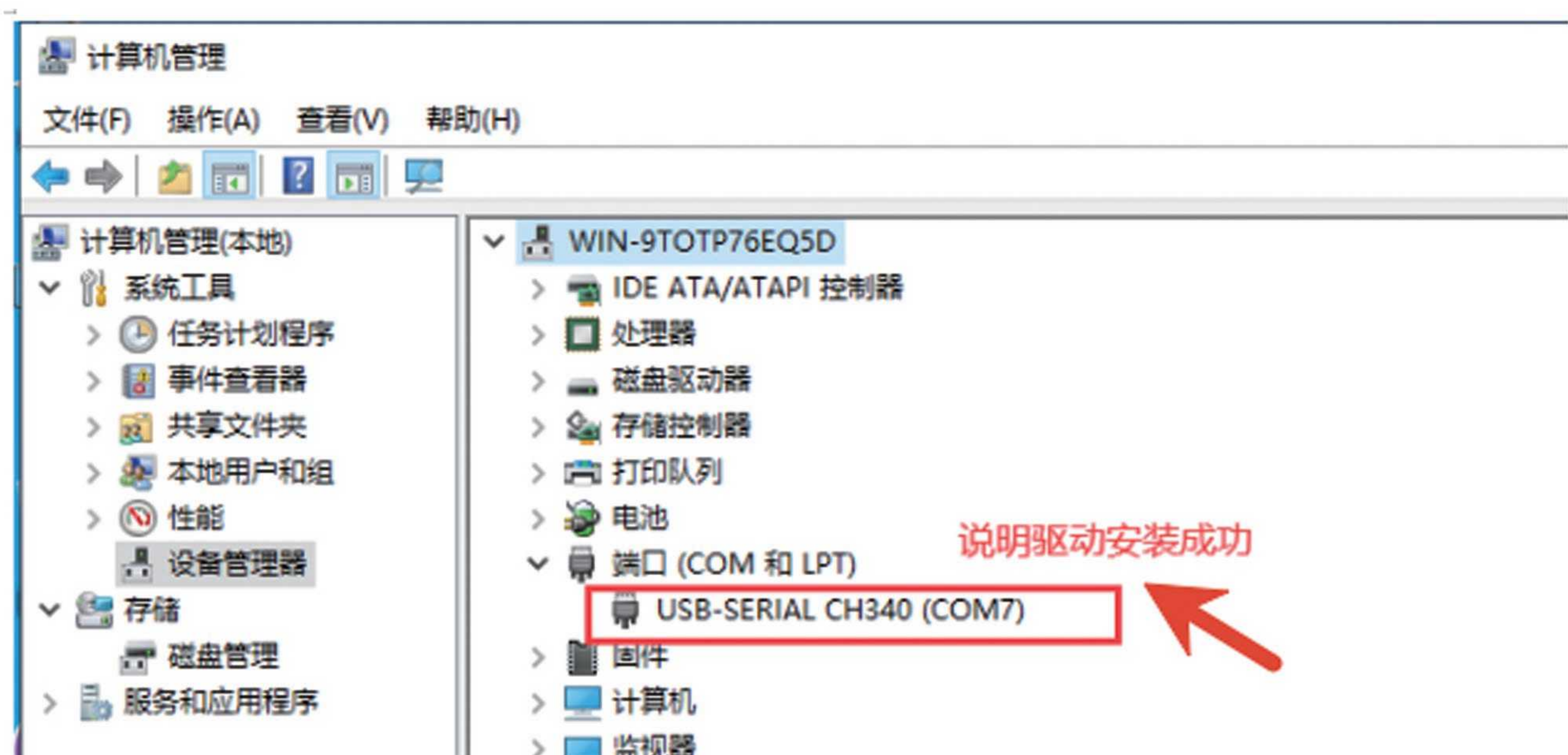
- 1、 Installation of serial port driver for development board
1. Open the provided 06 development board serial port driver CH343, double-click the driver, and click "Install" to start the installation

此电脑 > gy (G:) > ESP32-S3教程 > 核心板 > ESP32-S3-核心板 > 06-开发板串口驱动CH343 > Windows			
名称	修改日期	类型	大小
 CH343SER.EXE	2024/10/15 星期二 0:...	应用程序	608 KB

2. The following is the successful installation



3. Connect the development board to the computer using a USB cable



2、 Installation of Arduino development environment

1. Double click the Arduino installation package to install Arduino:

名称	修改日期	类型	大小
arduino-1.8.13-windows.exe	2024/10/15 星期二 0:...	应用程序安装包	114,267 KB
esp32-s3.zip	2024/10/16 星期三 0:...	360压缩 ZIP 文件	2,151,556...

2. Code package path setting:



3. Code directory structure:

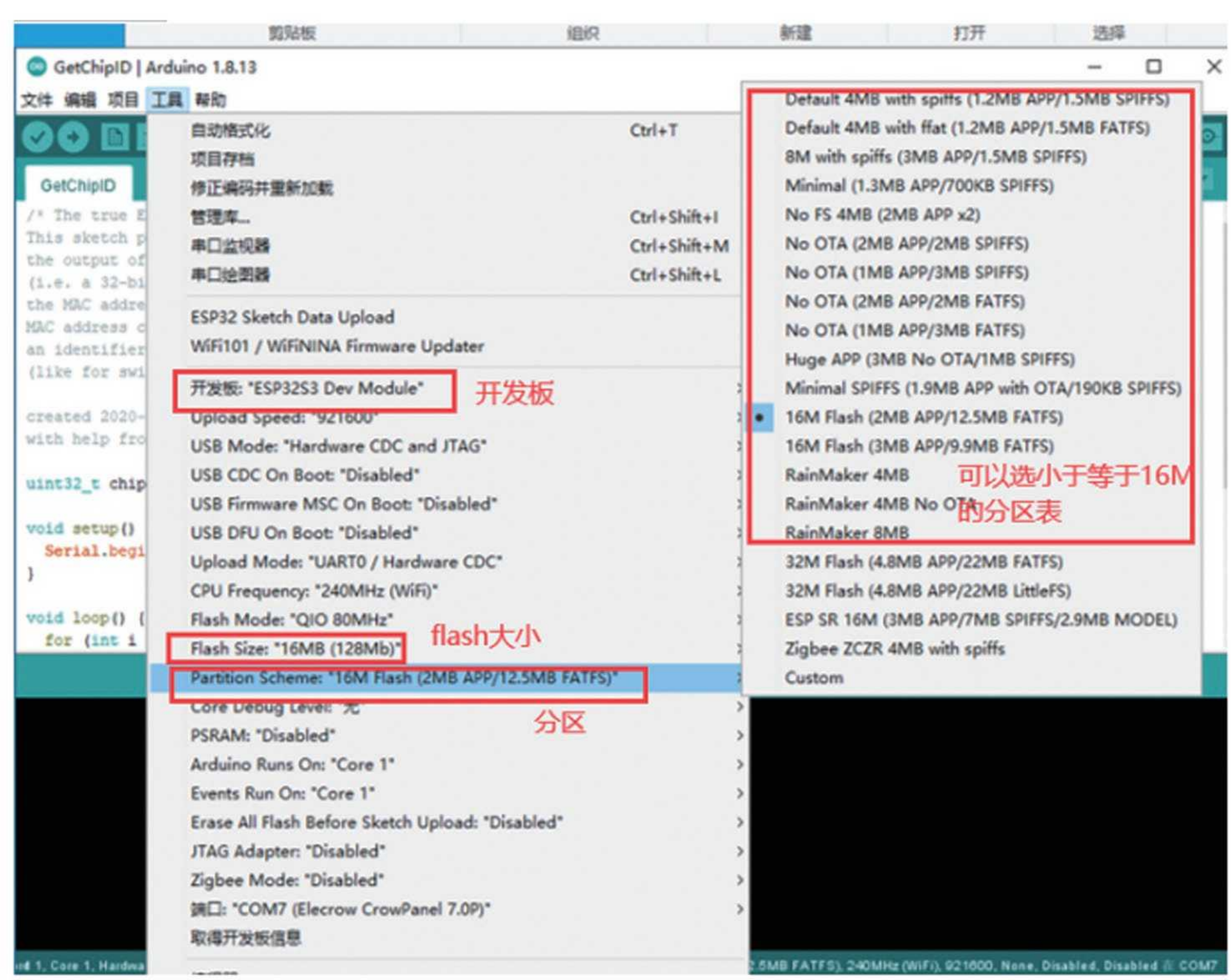


be careful:

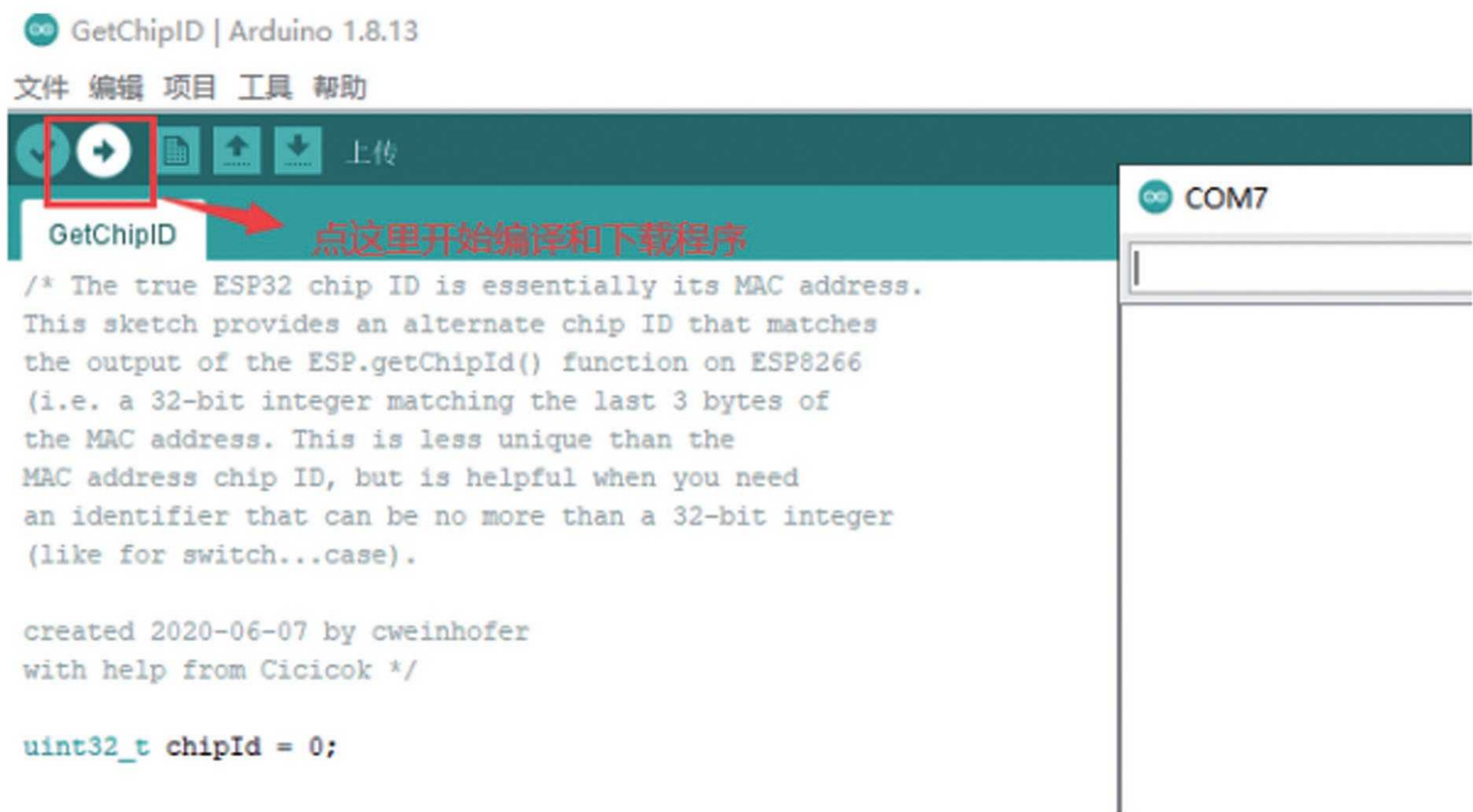
- 1) Be sure to manually create the hardware directory
- 2) Extract the code directly to the hardware directory
- 3) There are two ESP32 directories

After setting the path, be sure to restart Arduino!!!!

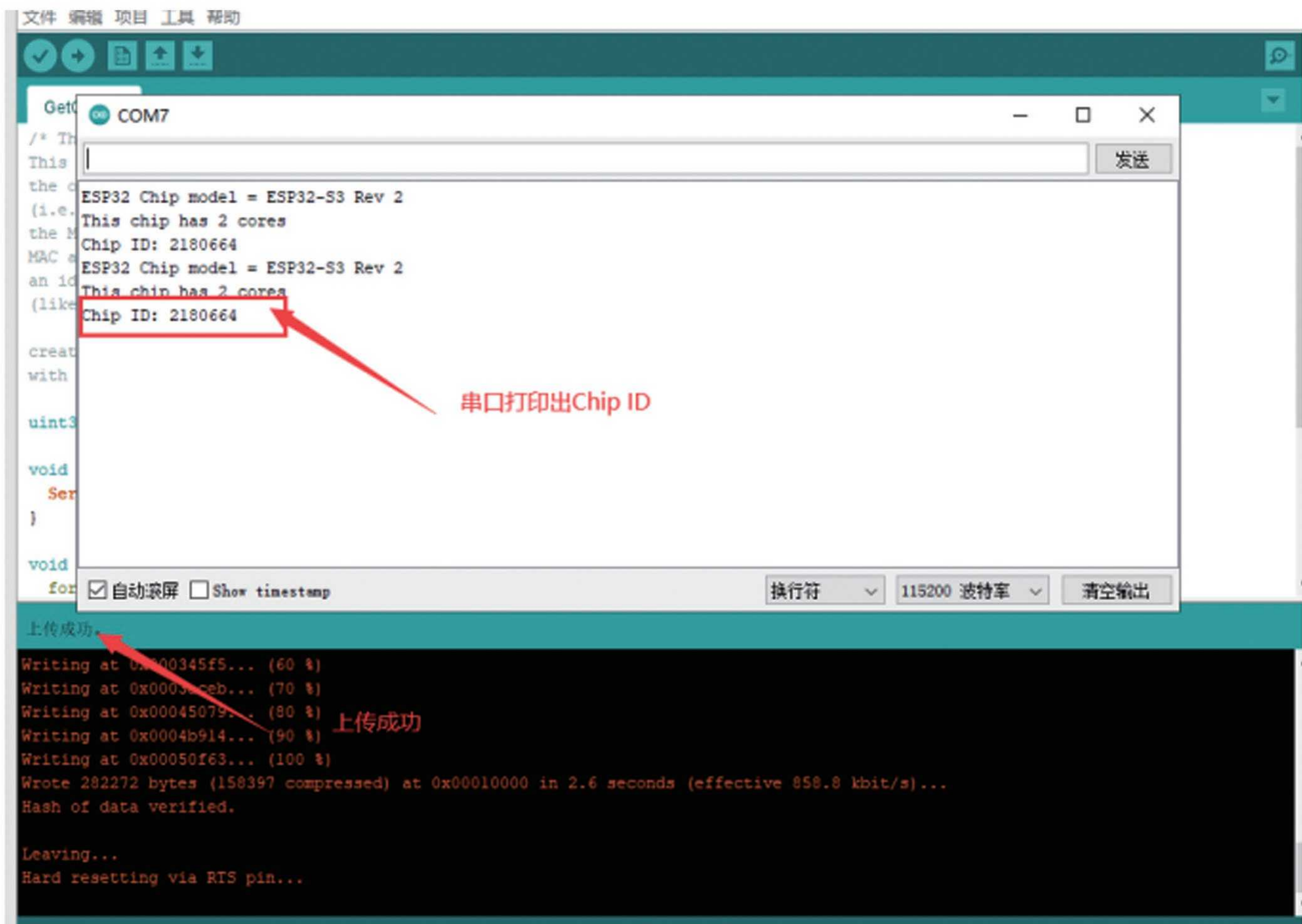
4. Open a routine below, select the development board, serial port, and partition for verification, and then click "Upload" to verify that the environment has been installed:



5. Download verification:



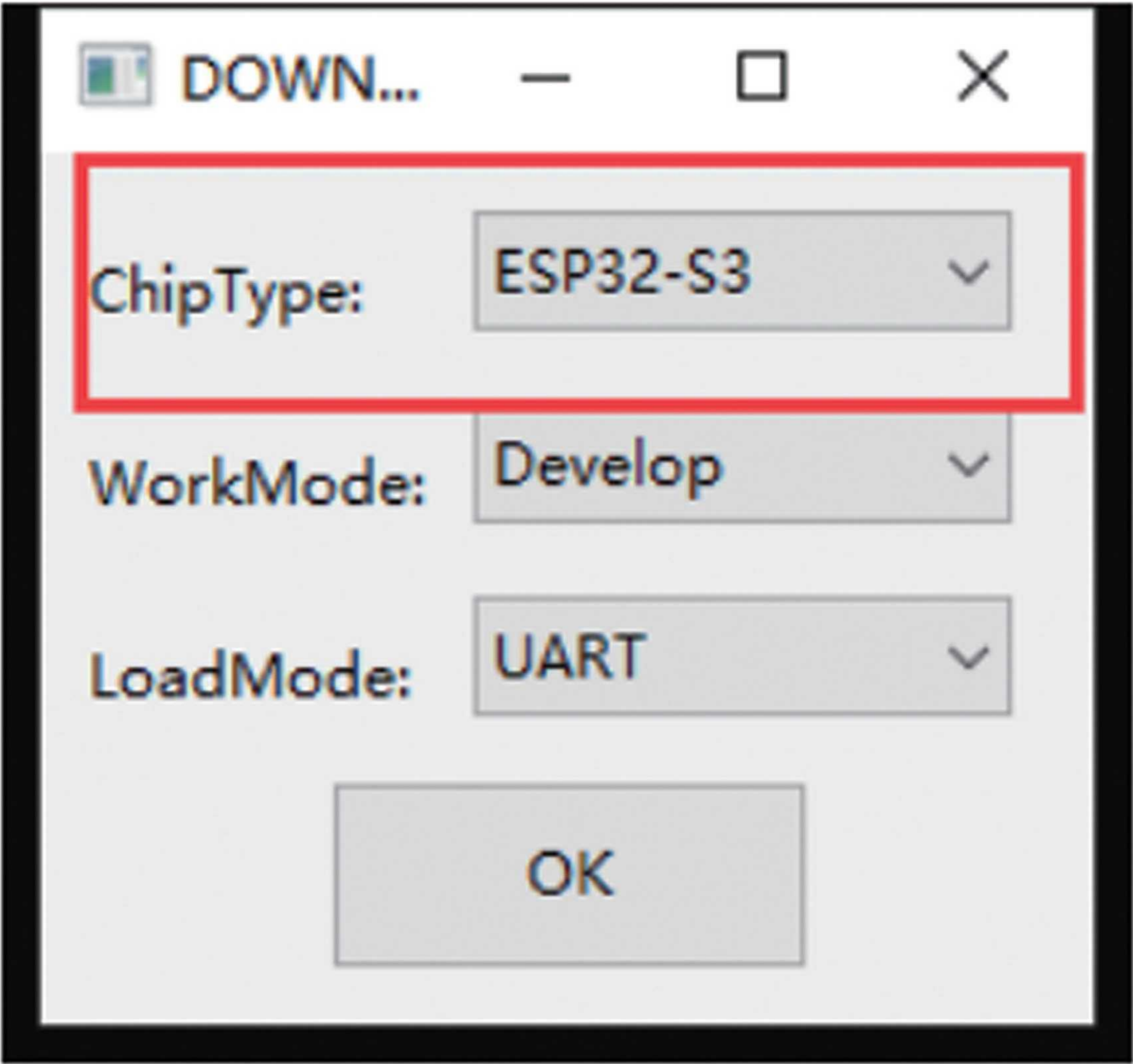
6. It can be seen through the "serial monitor" that chipID is being played, indicating that the environment is ready:



3、 Installation of Micropython Development Environment

ESP32-S3-N16R8-MPY-V1.1.bin	2024/10/15 星期二 0:...	BIN 文件	1,385 KB
flash_download_tool_3.9.4_0.zip	2023/7/22 星期六 22:...	360压缩 ZIP 文件	16,412 KB
test-main.py	2024/10/15 星期二 0:...	Python File	1 KB
thonny-4.0.2.exe	2024/10/15 星期二 0:...	应用程序	21,006 KB

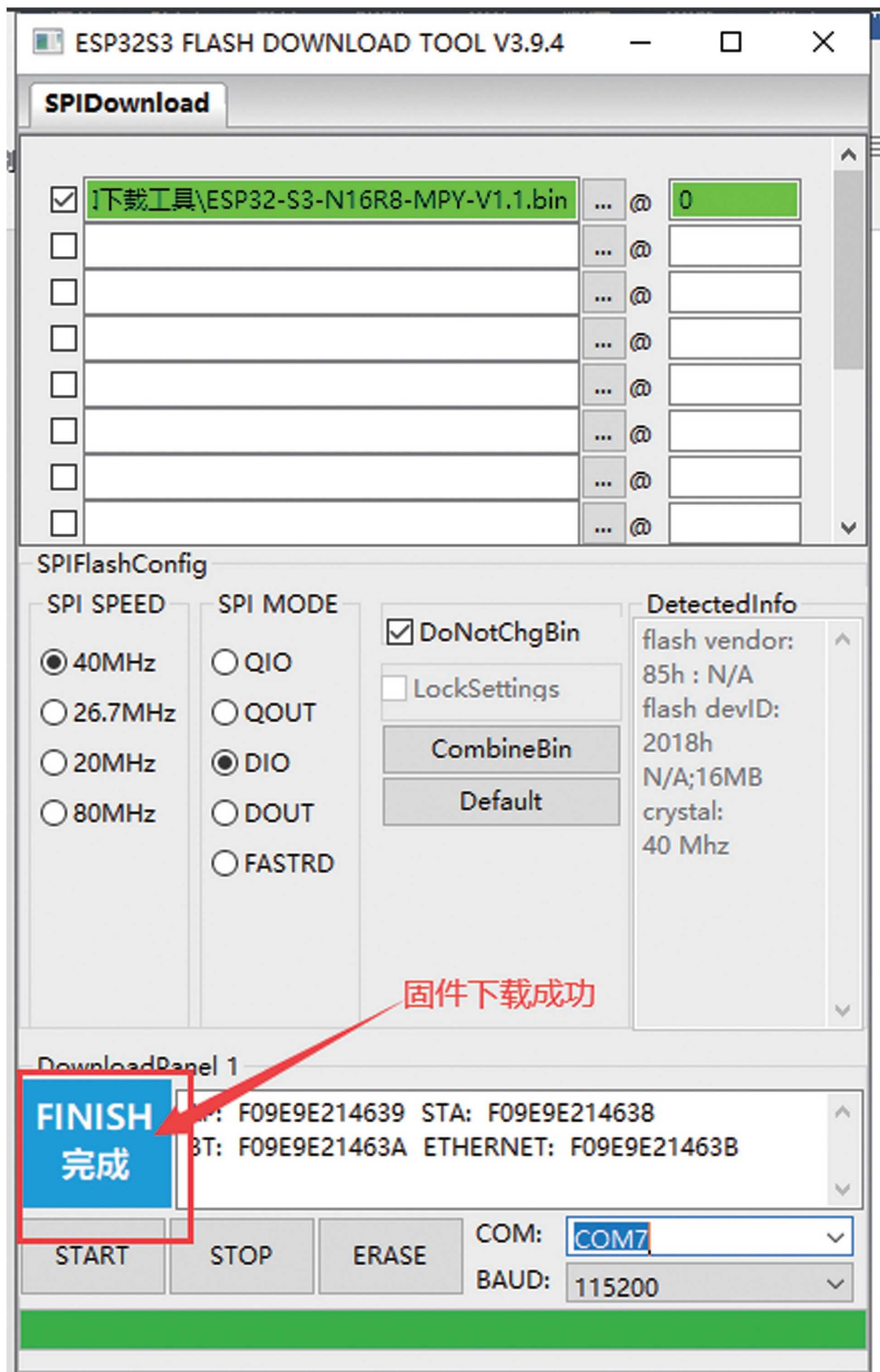
1. Extract the micropython download toolkit, double-click to open the micropython firmware download tool, and select "ESP32-S3"



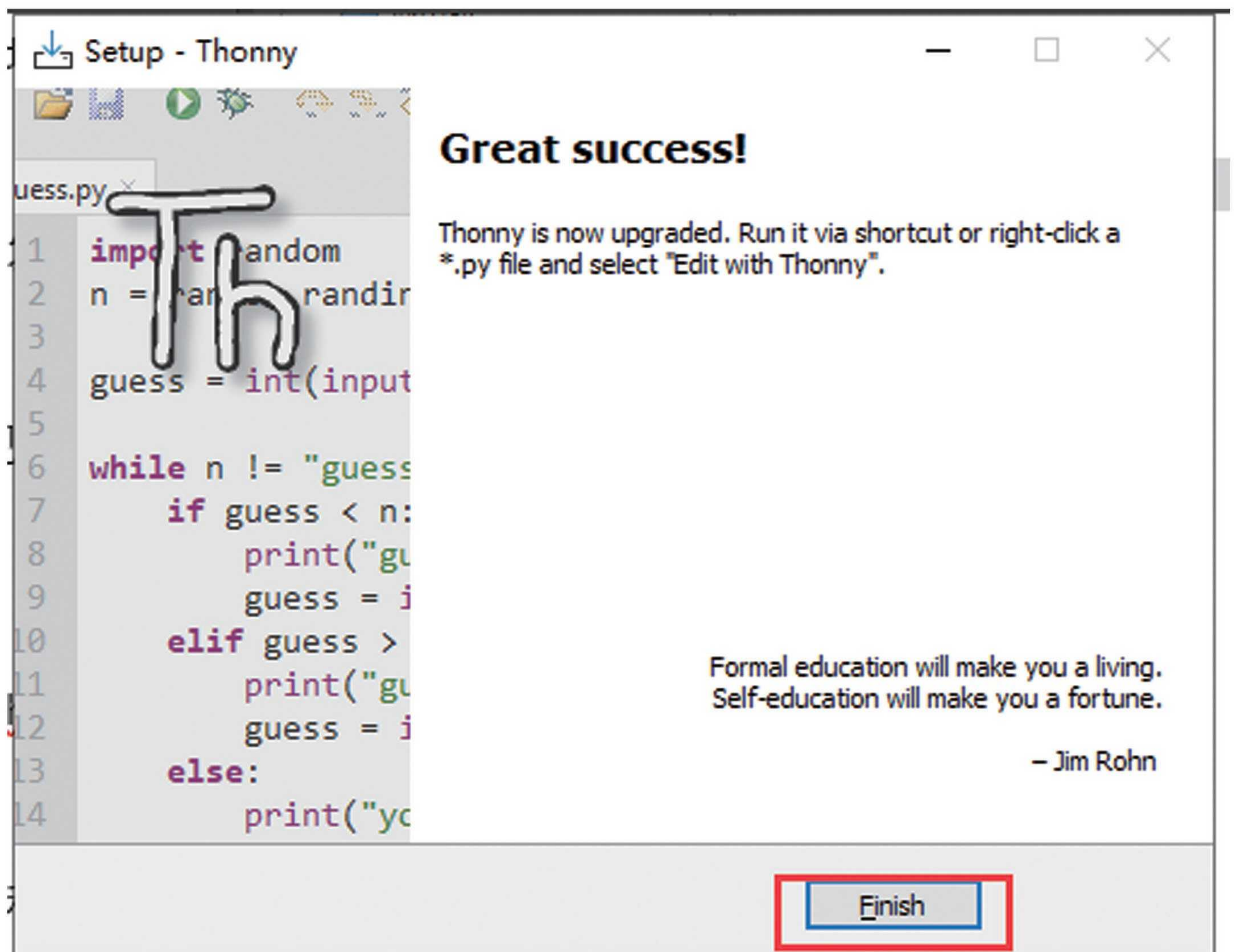
2. Select firmware to start downloading:



3. Click "START" to start downloading:



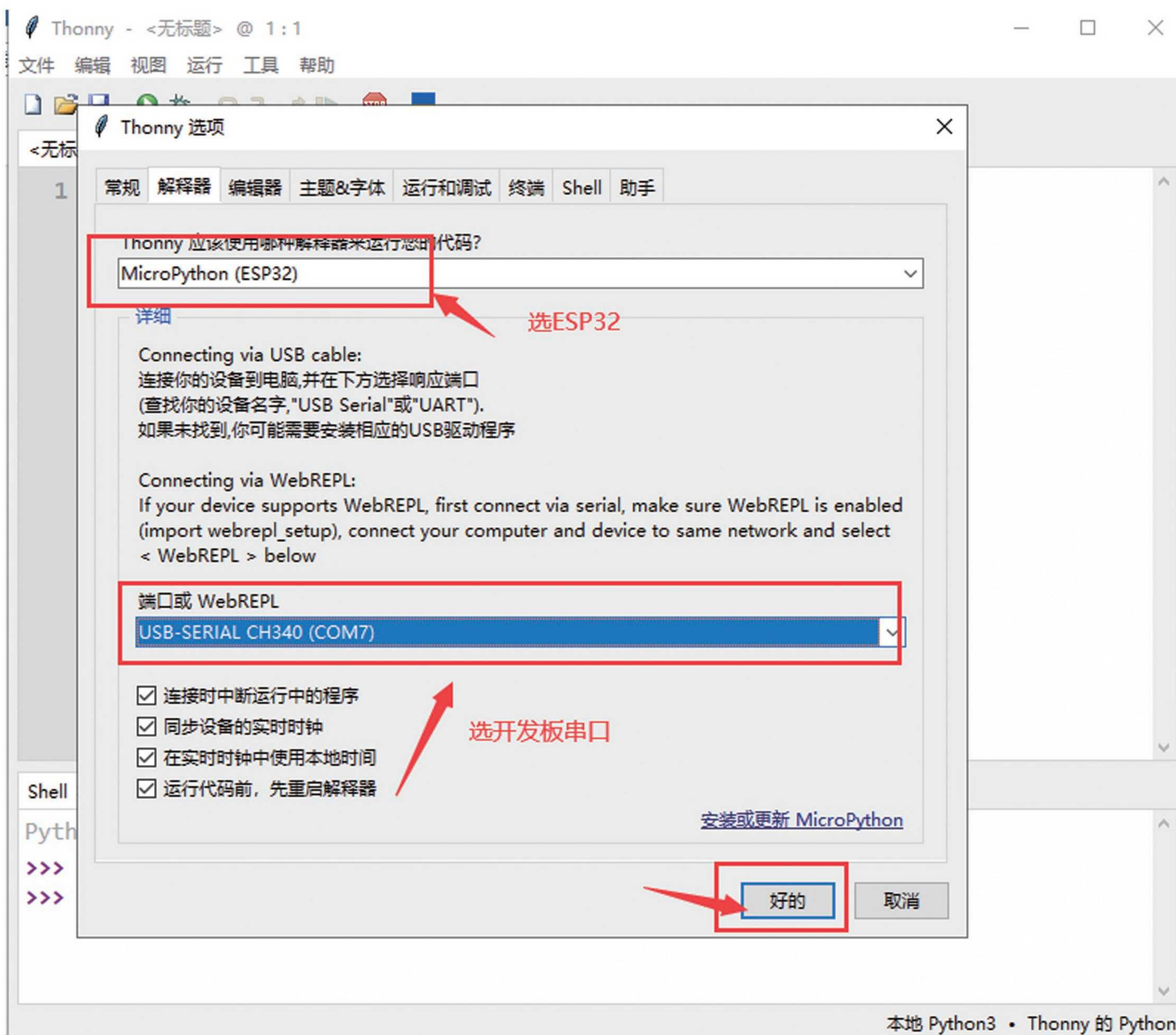
4. Double click to install thongy-4.0.2.exe, the next step is to complete the installation:



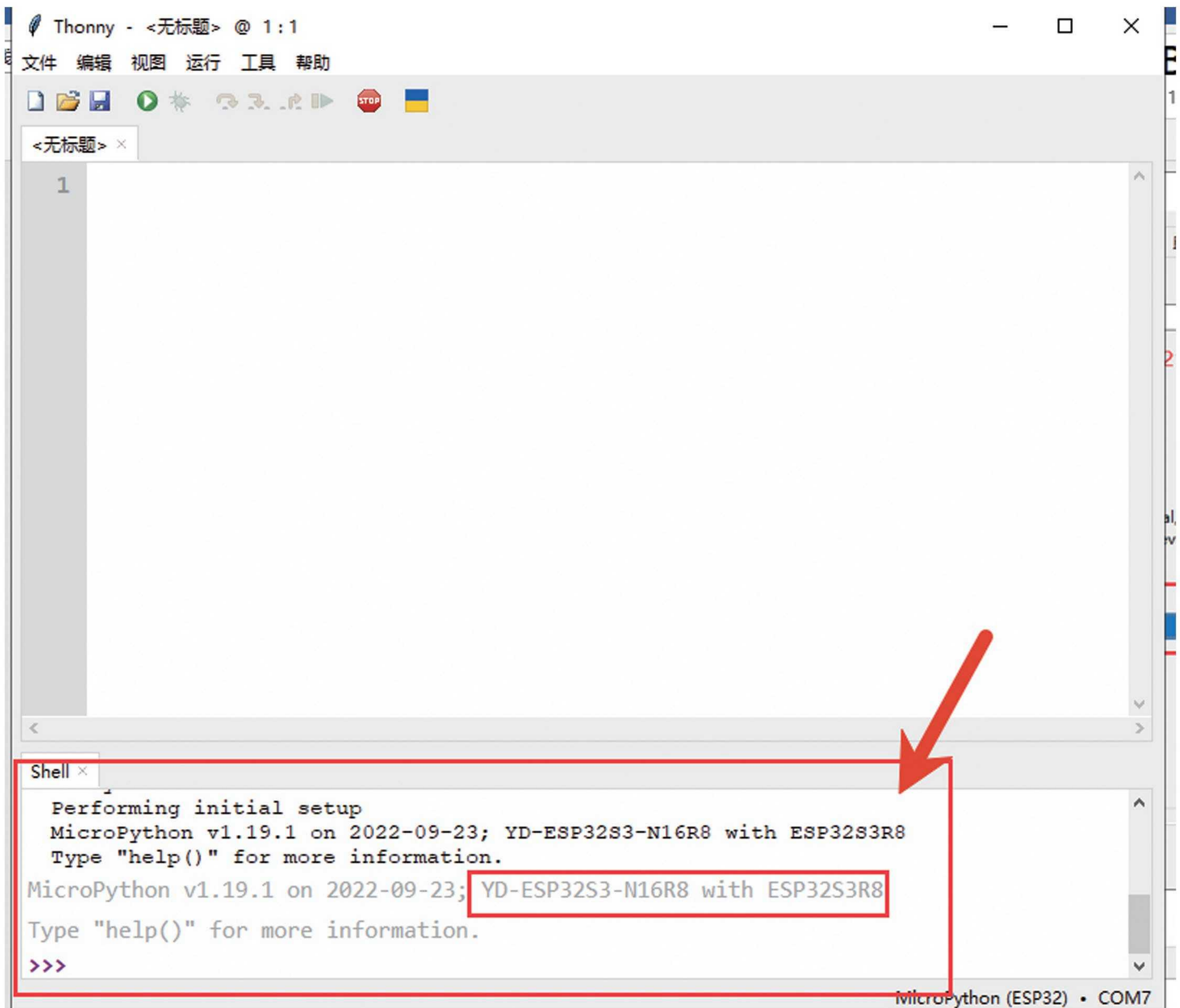
5. Open Python for environment validation, select the language option:



6. Select configuration, "Run" -> "Configure Interpreter"



Press the RST key on the development board, and you will see that Python has already connected to the board,



Thonny - <无标题> @ 1:1

文件 编辑 视图 运行 工具 帮助

<无标题> x

1

Shell x

```
Performing initial setup
MicroPython v1.19.1 on 2022-09-23; YD-ESP32S3-N16R8 with ESP32S3R8
Type "help()" for more information.
MicroPython v1.19.1 on 2022-09-23; YD-ESP32S3-N16R8 with ESP32S3R8
Type "help()" for more information.
>>>
```

MicroPython (ESP32) • COM7

7. Open a test history test main.cy to conduct testing

Thonny - G:\ESP32-S3教程\核心板\ESP32-S3-核心板\03-micropython教程和资料\micropython固件和编程工具和下载工具\test-main.py @ 12:15

文件 编辑 视图 运行 工具 帮助

1、点“运行”，测试例程就运行起来了

```
5 pin = Pin(48, Pin.OUT)
6 np = NeoPixel(pin, 1)
7 np[0] = (10,0,0)
8 np.write()
9 r, g, b = np[0]
10 def handle_interrupt(Pin):
11     np[0] = (0, 0, 0)
12     np.write()
13     time.sleep_ms(150)
14
15     np[0] = (0, 0, 10)
16     np.write()
17     time.sleep_ms(150)
18
19     np[0] = (0, 0, 10)
20     np.write()
21     time.sleep_ms(150)
22
23     np[0] = (0, 0, 0)
24     np.write()
25     time.sleep_ms(150)
26
27     np[0] = (0, 10, 0)
28     np.write()
29     time.sleep_ms(150)
30
31     print("test-usr key")
32 p0 = Pin(0)
33 p0.init(p0.IN, p0.PULL_UP)
34 p0.irq(trigger=p0.IRQ_FALLING, handler=handle_interrupt)
```

Shell x

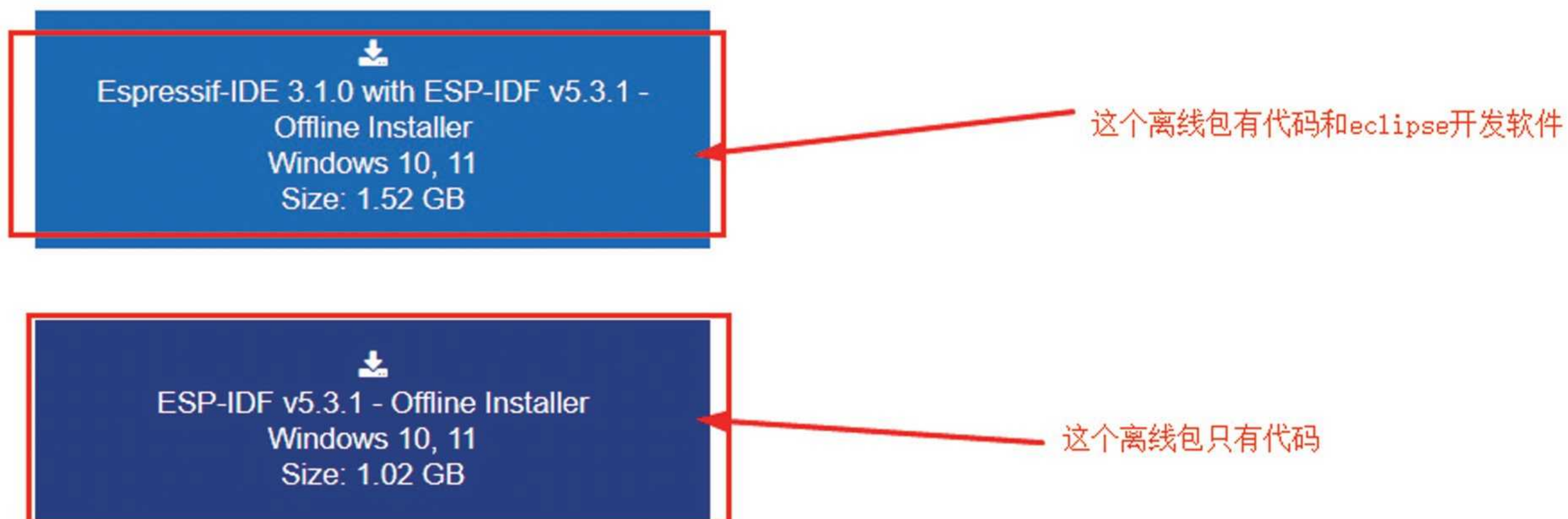
```
test-usr key
test-usr key
test-usr key
test-usr key
test-usr key
test-usr key
```

2、按开发板的boot键，可以看到打印，并且开发板上的led灯会发生颜色改变

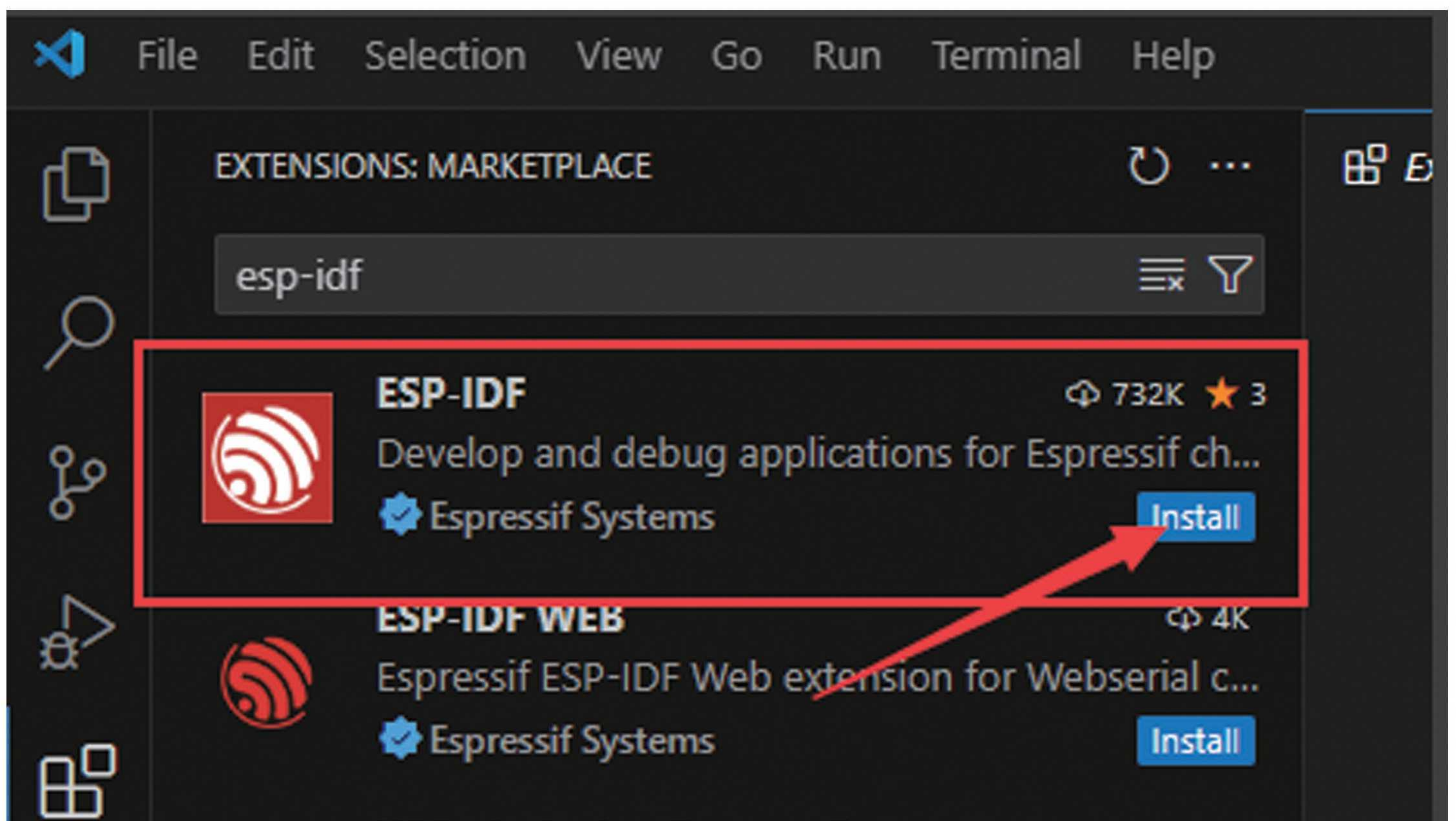
At this point, the micropython development environment has been set up.

4、 Installation of Idf development environment

1. Idf offline package download address: <https://dl.espressif.cn/dl/esp-idf/?idf=4.4> You can download offline packages from this address



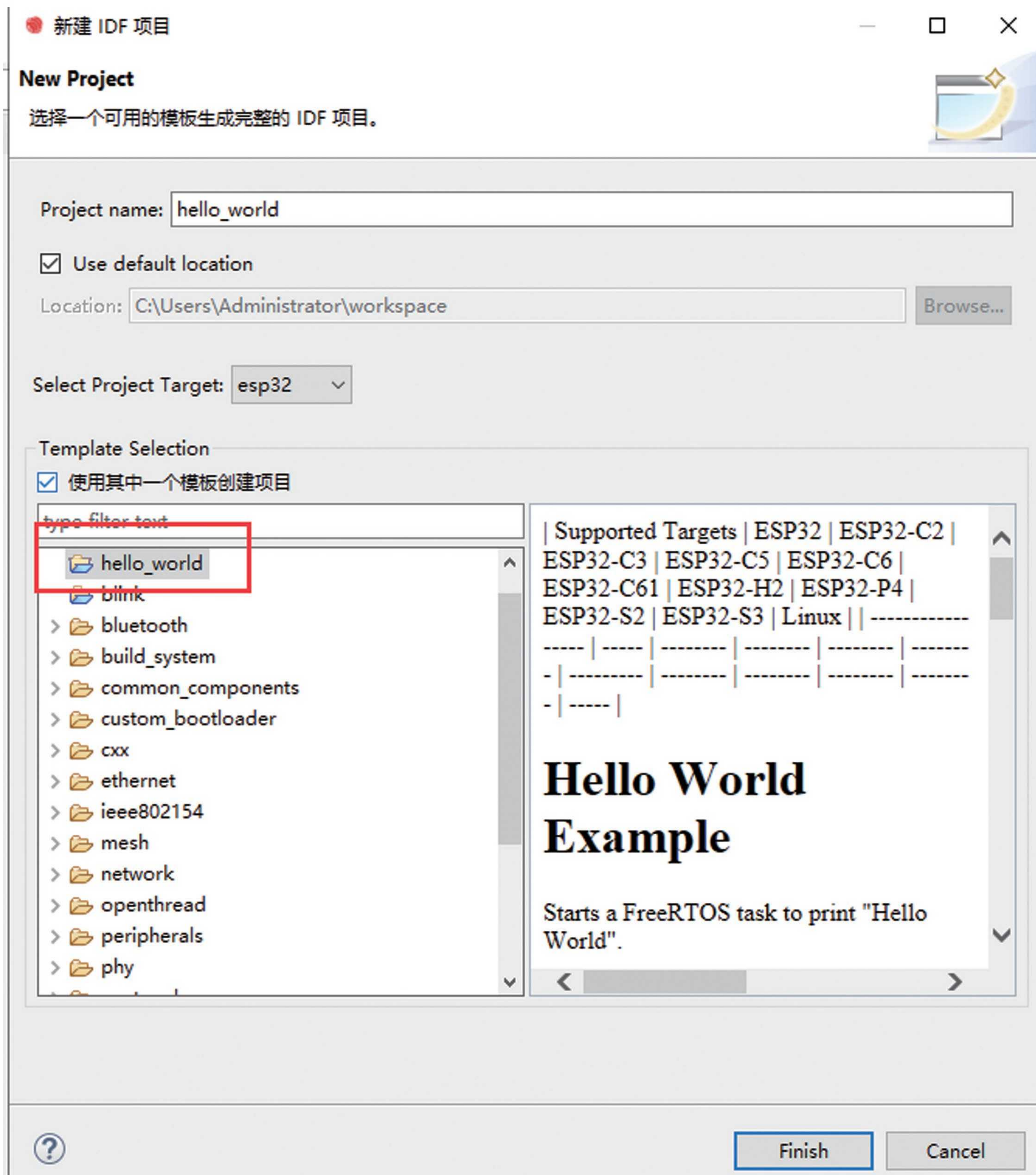
2. Install ESP-IDF plugin on VSCode



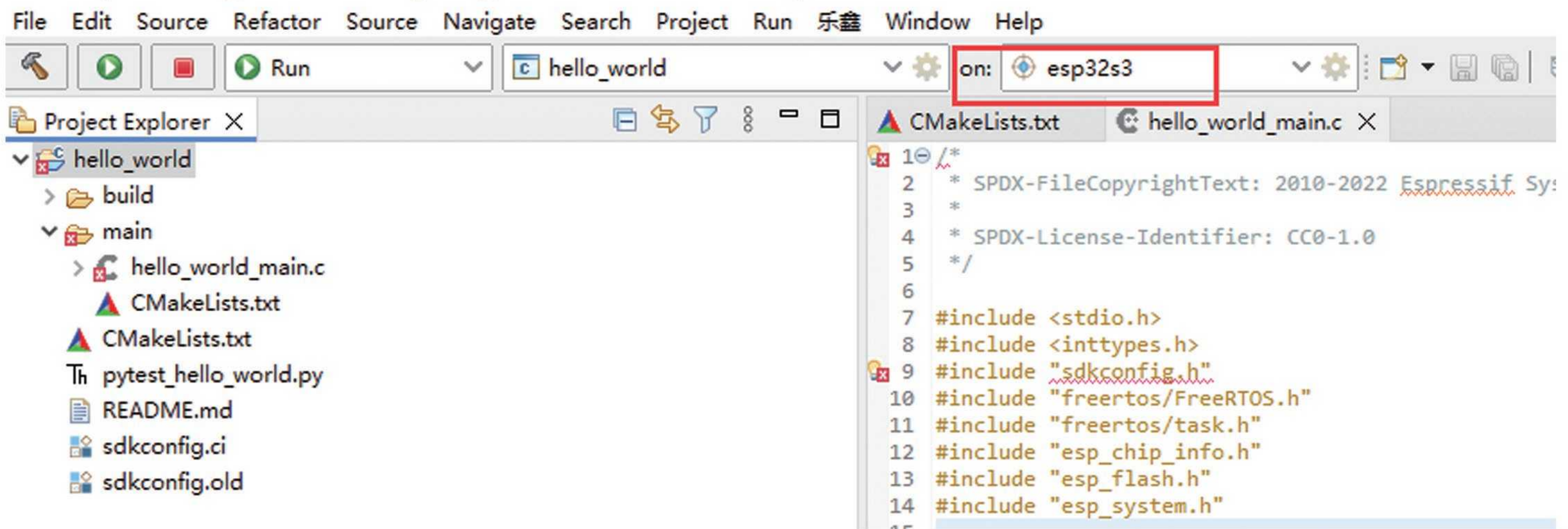
Next, let's watch the video tutorial on how to configure and solve the problems encountered.

1) Developing with Eclipse

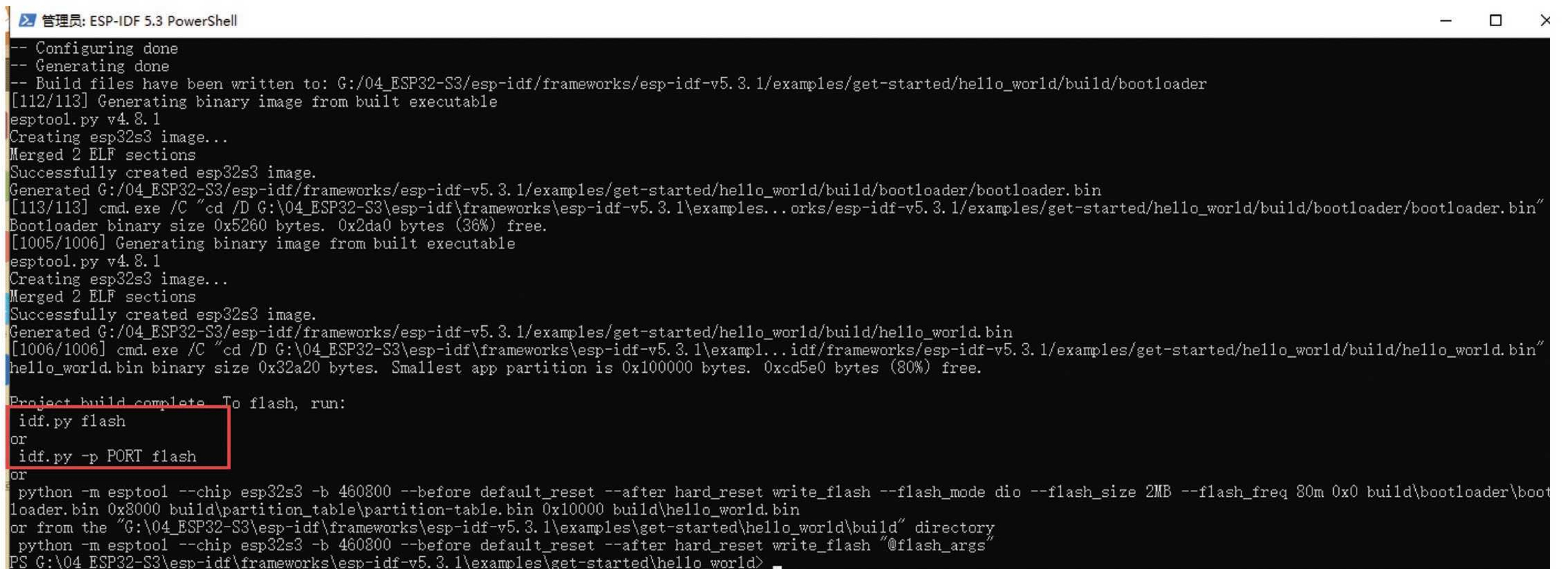
If the installation of the Python package is slow because it is installed using foreign Python sources by default, you can search online to set up domestic Python sources.



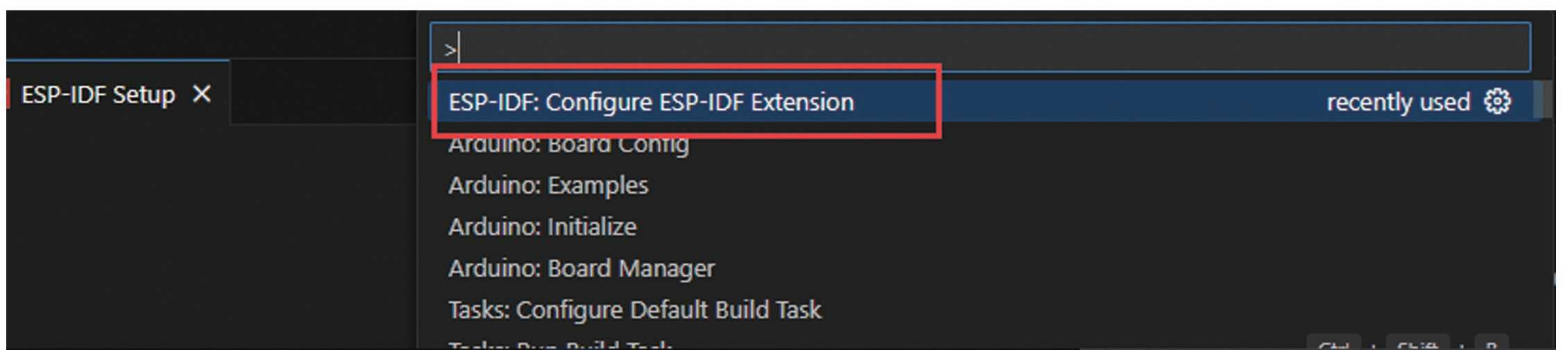
Set the compilation target to esp32s3

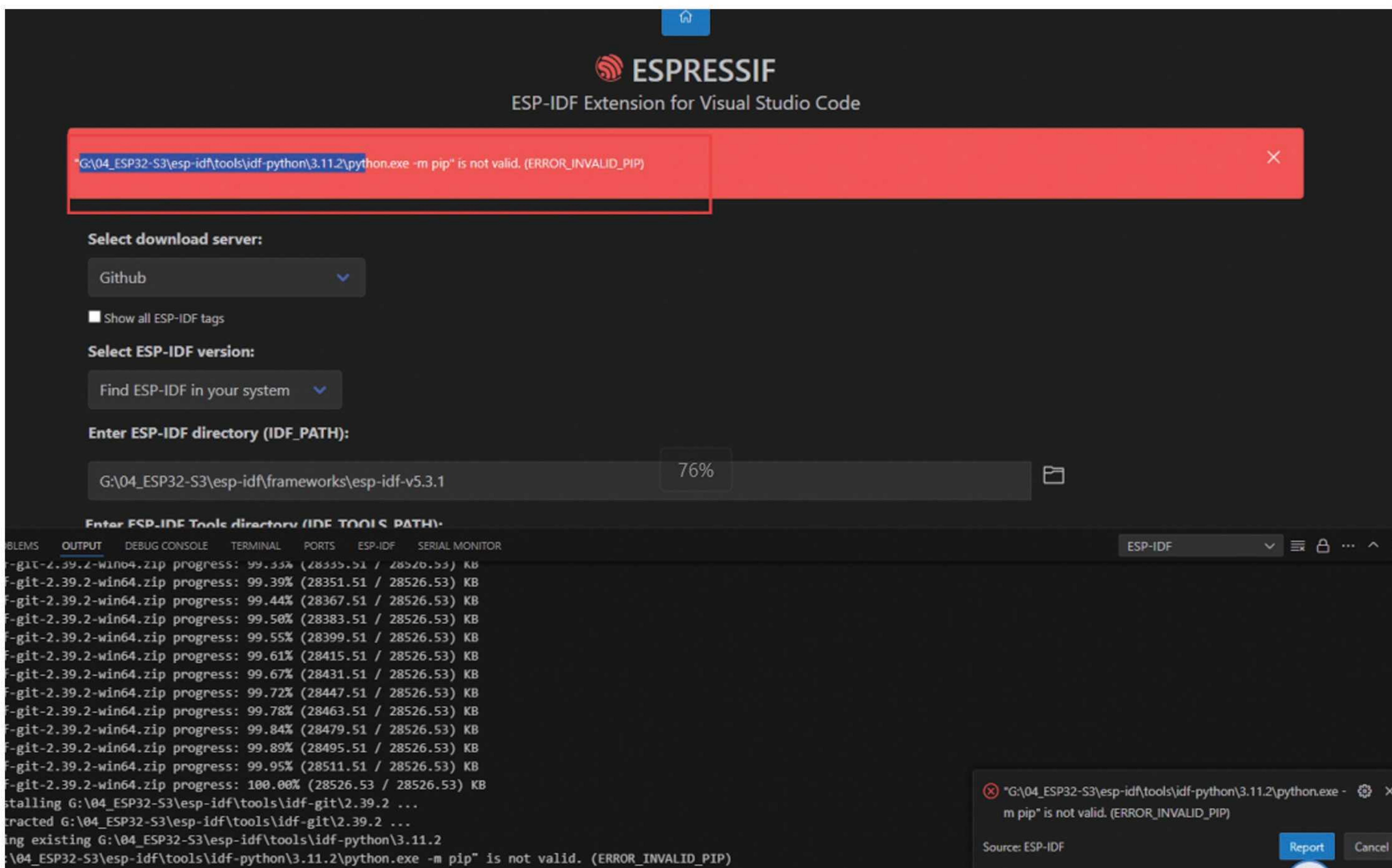
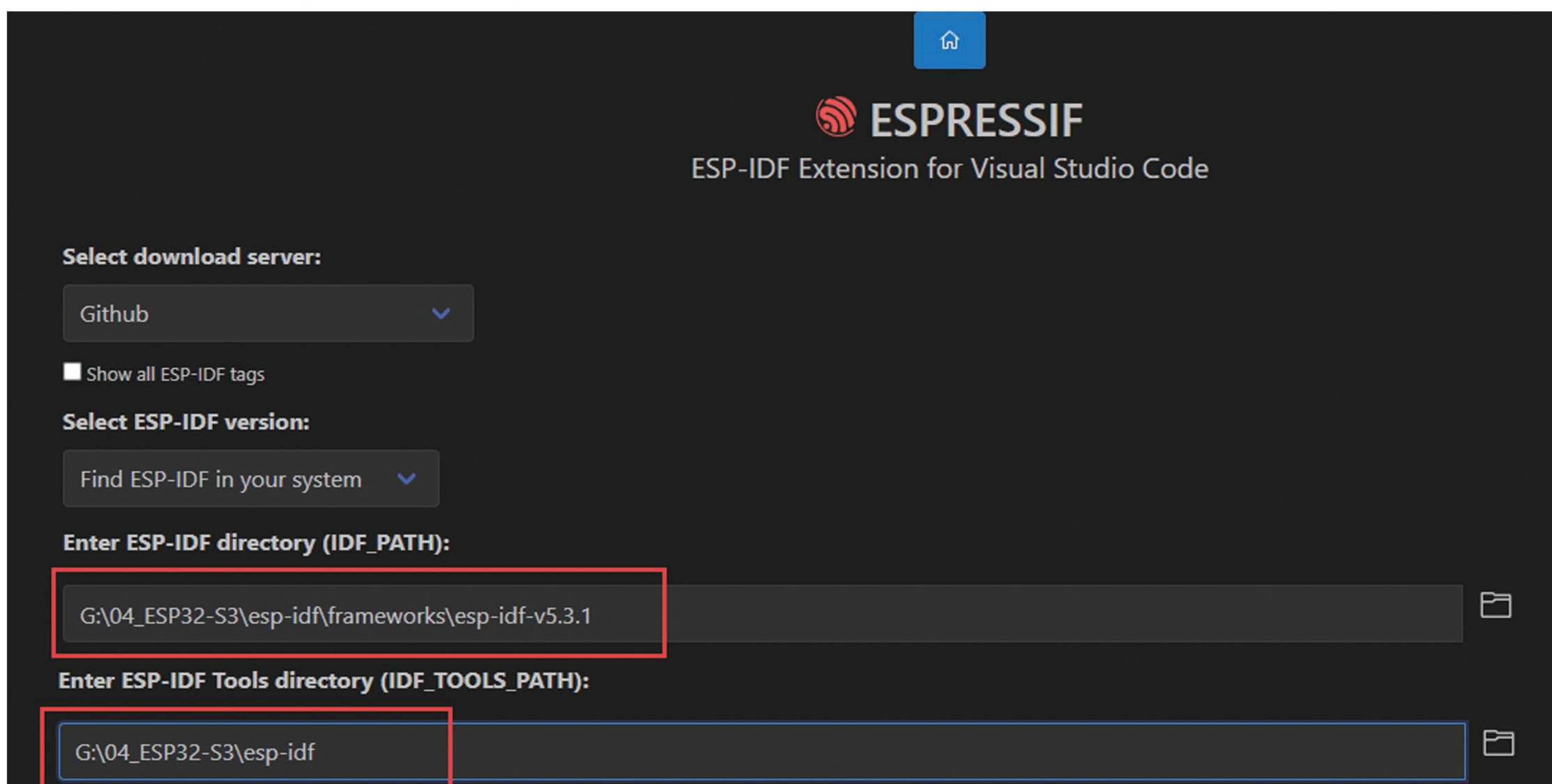


2) Develop using the Windows command line

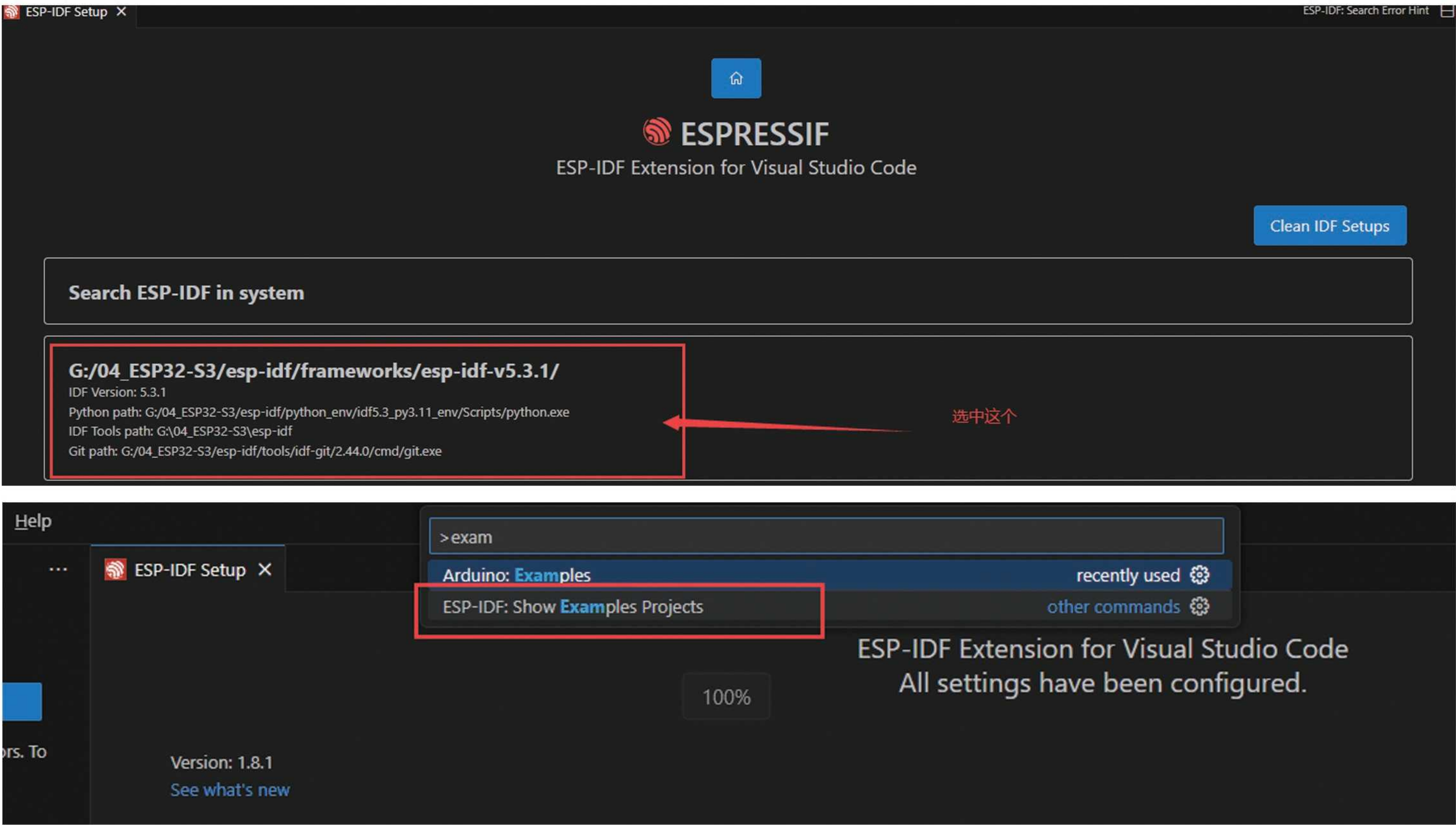


3) Developing with VSCode (highly recommended)

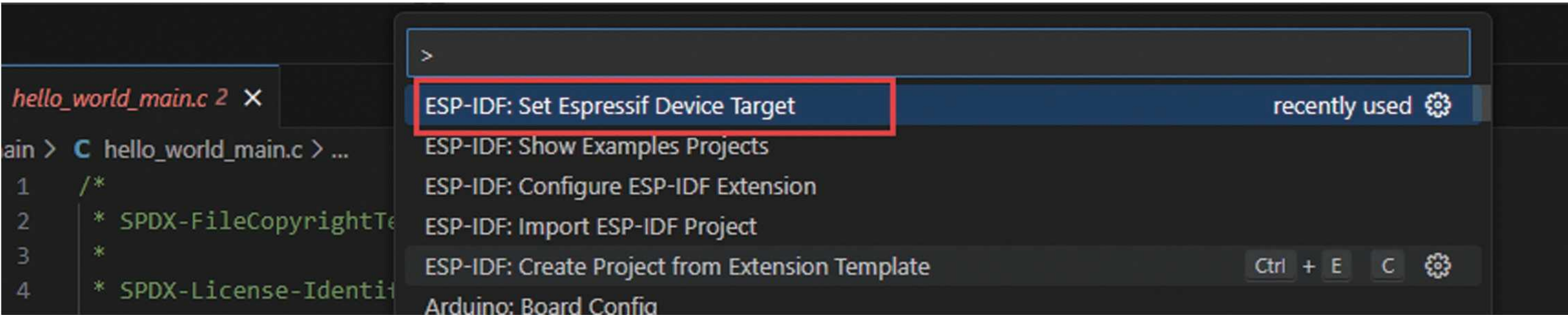




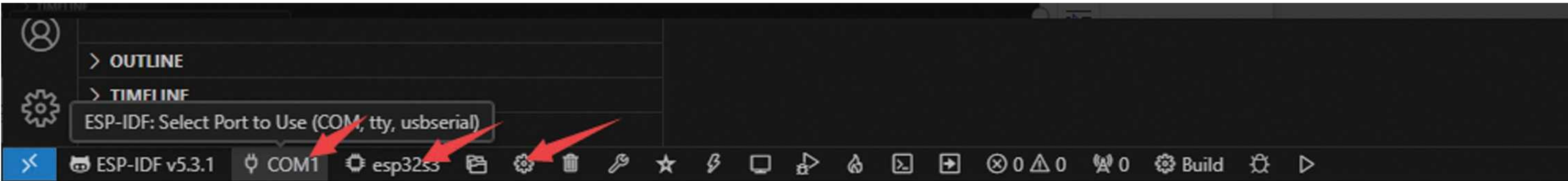
Solution: Restart VSCode and select the configuration again to solve the problem.



Configure development board:



There is a quick configuration tool in the bottom left corner:



Compilation completed:

28
29

(chip_info.features & CHIP_FEATURE_BT) ? "BT" : "",
(chip_info.features & CHIP_FEATURE_BLE) ? "BLE" : "",

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

ESP-IDF

SERIAL MONITOR

ESP-IDF

[/SDK Configuration Editor]
[Build]
Project build complete. To flash, run:
ESP-IDF: Flash your project in the ESP-IDF Visual Studio Code Extension
or in a ESP-IDF Terminal:
idf.py flash
or
idf.py -p PORT flash
or
python -m esptool --chip esp32s3 -b 460800 --before default_reset --after hard_reset --port COM7 write_flash --flash_mode dio --flash_size 2MB --flash_freq 80m 0x0 bootloader/bootloader
bin 0x10000 hello_world.bin 0x8000 partition_table/partition-table.bin
or from the "g:\04_ESP32-S3\exa\hello_world\build" directory
python -m esptool --chip esp32s3 -b 460800 --before default_reset --after hard_reset write_flash "@flash_args"

Build Successfully

5、 Vscode development environment setup

After installing the Arduino or IDF development environment, VSCode will only work

1) Double click the VSCode installation package to install VSCode,

05-串口助手

06-开发板串口驱动CH343

07_蓝牙调试助手

08-串口调试助手和网络调试助手

VSCodeUserSetup-x64-1.94.2.exe

必看-资料介绍和开发板驱动安装.mp4

教程文档.docx

2024/10/10 星期一 1:...

2024/10/15 星期二 0:...

2024/10/15 星期二 0:...

2024/10/16 星期三 1:...

2024/10/16 星期三 1:...

2024/10/16 星期三 1:...

2024/10/16 星期三 2:...

文件夹

文件夹

文件夹

文件夹

应用程序

MP4 文件

Microsoft Word ...

98,041 KB

35,504 KB

1,433 KB

vscode安装包

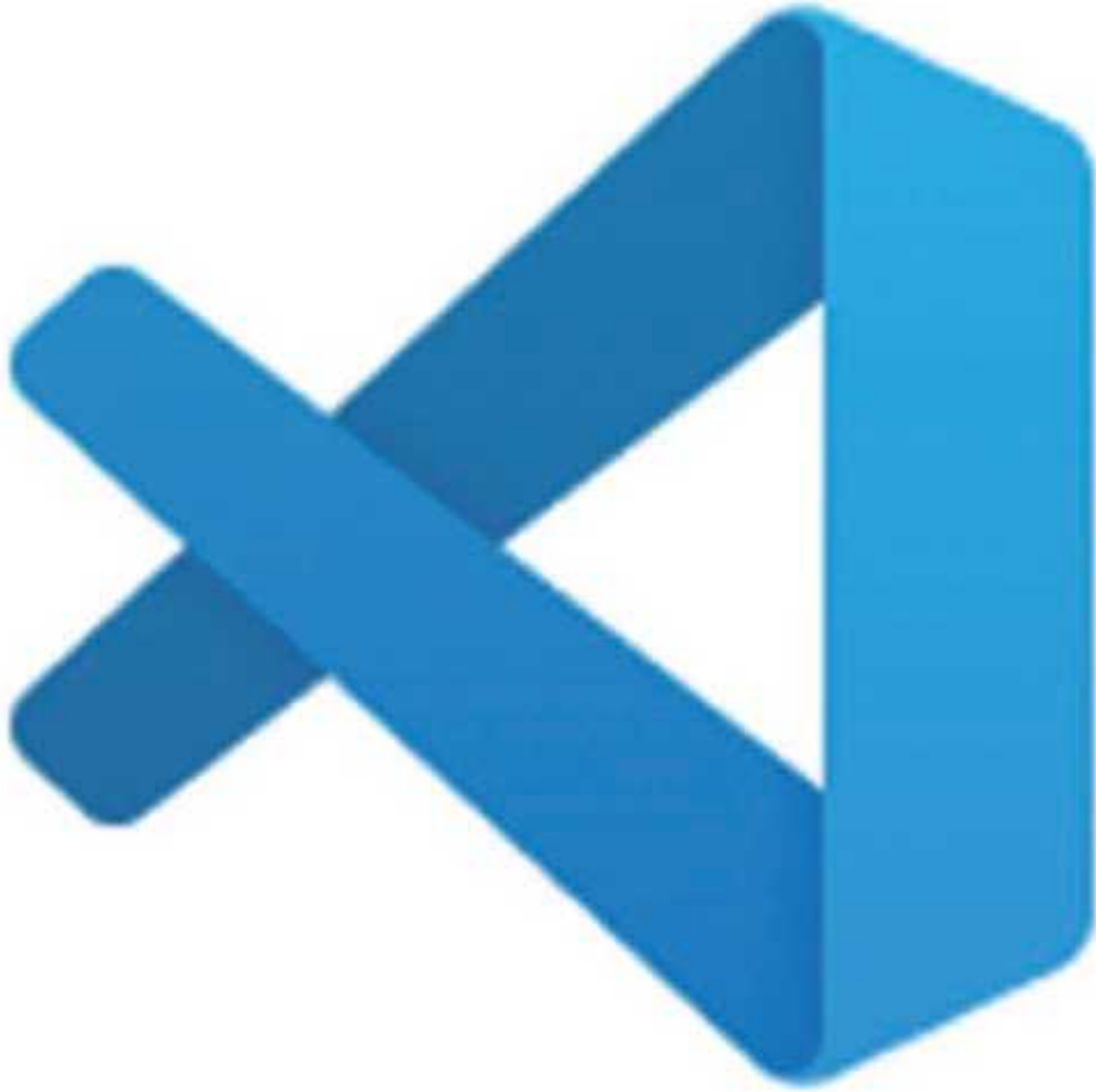
安装 - Microsoft Visual Studio Code (User)

Visual Studio Code 安装完成

安装程序已在您的电脑中安装了 Visual Studio Code。此应用程序可以通过选择安装的快捷方式运行。

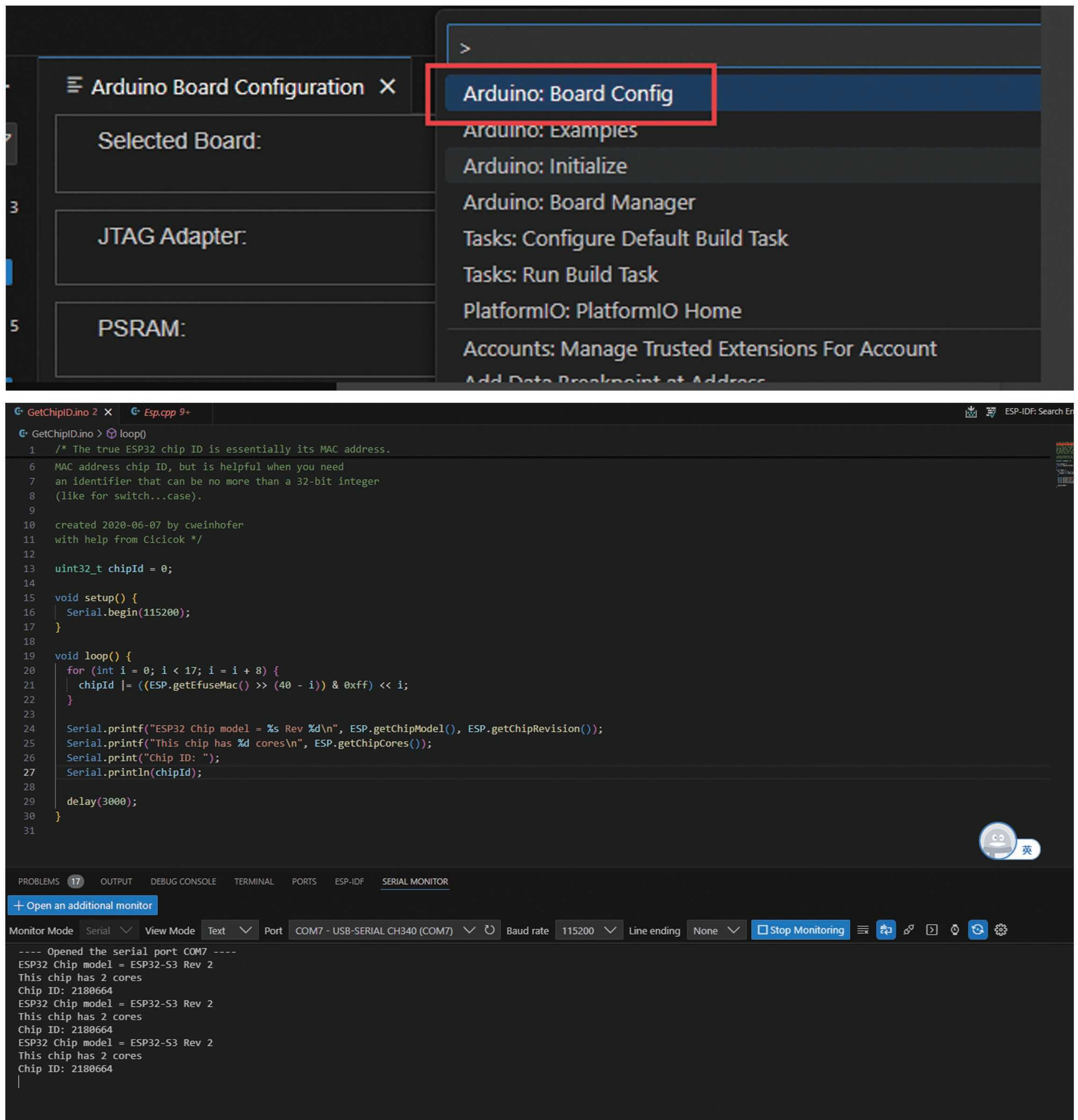
单击“完成”退出安装程序。

☒ 运行 Visual Studio Code

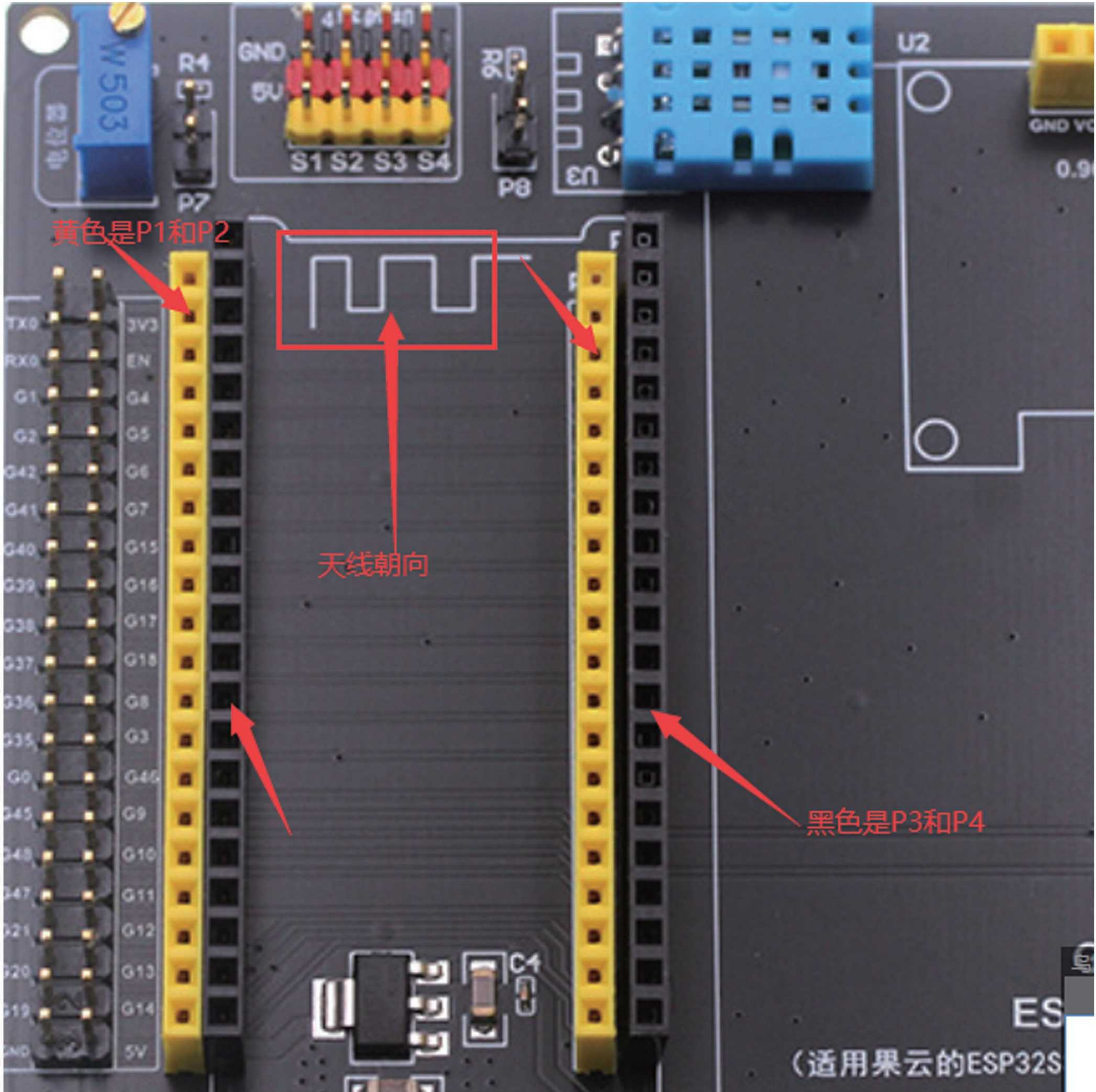
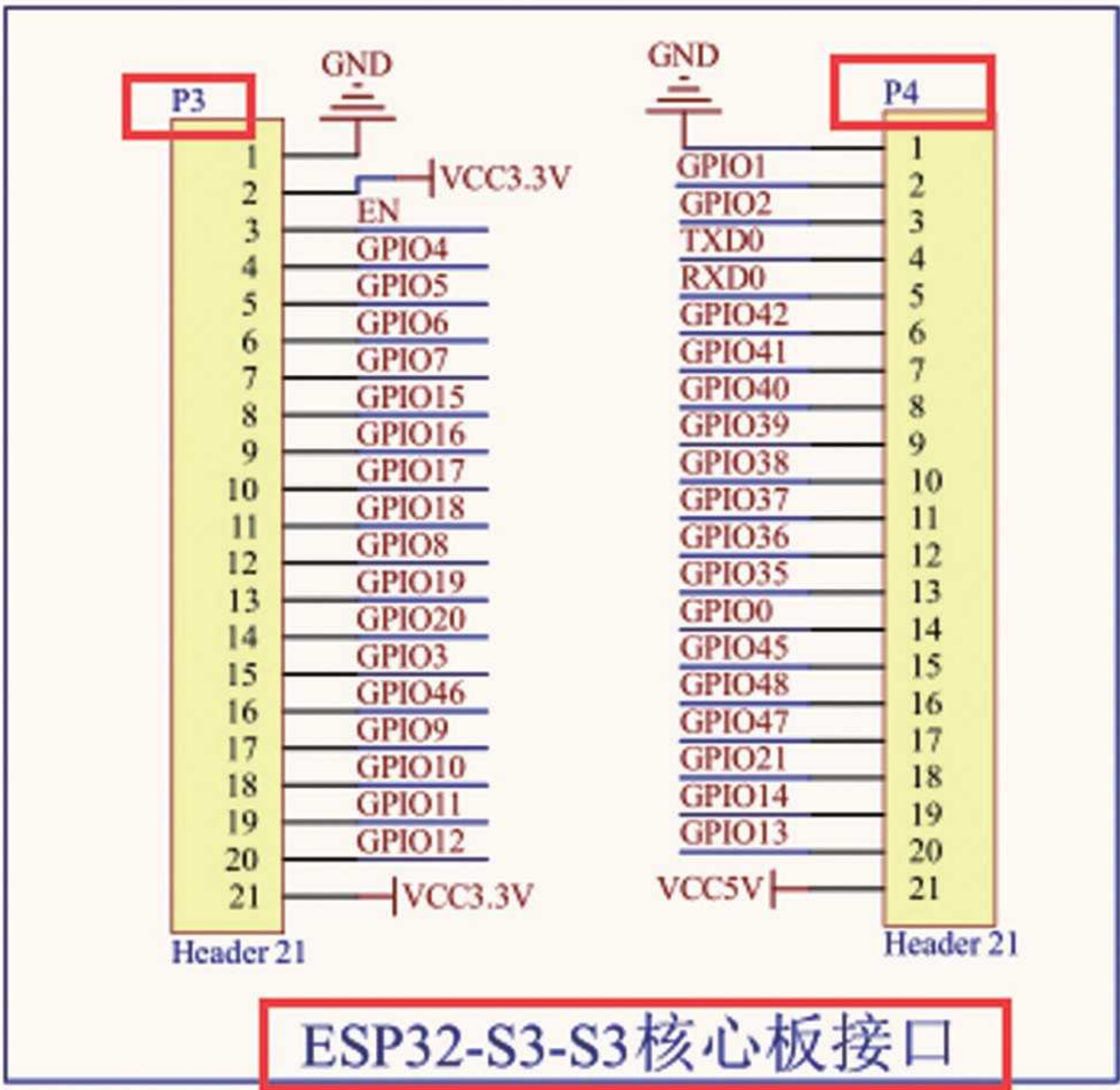
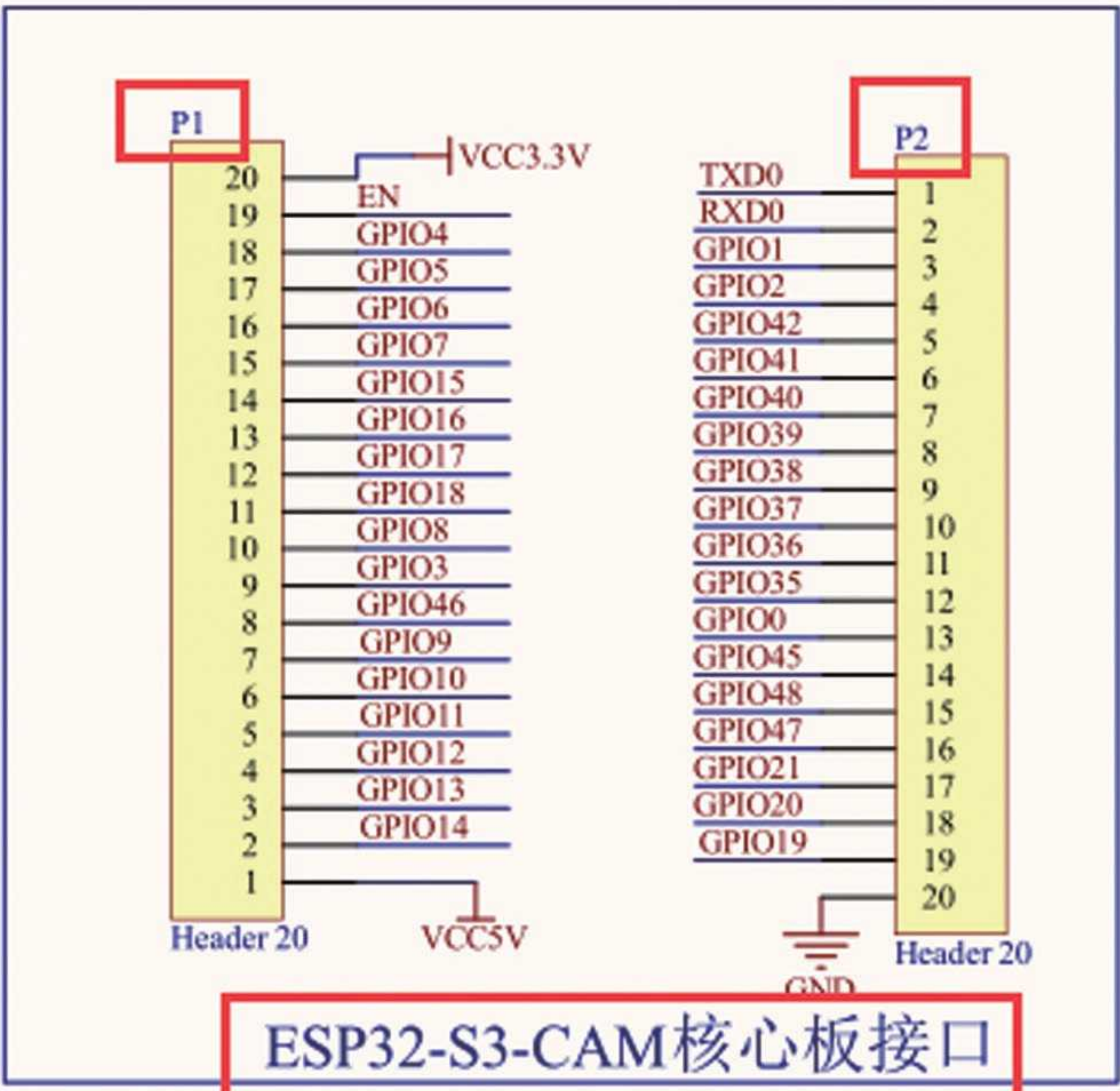


完成(F)

2) In the plugin market, install Arduino plugin or IDF plugin



Expansion board plug-in board instructions:



Future video tutorials will be based on the explanation of the expansion board.

1、 Arduino Video Tutorial

a) Copy the dependency library of the routine to the folder

比电脑 > gy (G:) > ESP32-S3教程 > 核心板 > ESP32-S3-核心板 > 04-arduino教程和资料 > 03-arduino例程和库 > libraries >

名称	修改日期	类型	大小
DHT11	2024/10/27 星期日 2...	文件夹	
ESP32Servo	2024/10/27 星期日 2...	文件夹	
ESP8266_and_ESP32_OLED_driver_for...	2024/10/27 星期日 2...	文件夹	
Freenove_WS2812_Lib_for_ESP32	2024/10/27 星期日 2...	文件夹	
lvgl	2024/10/27 星期日 2...	文件夹	
TFT_eSPI	2024/10/27 星期日 2...	文件夹	
TFT_Touch	2024/10/27 星期日 2...	文件夹	
U8g2	2024/10/27 星期日 2...	文件夹	

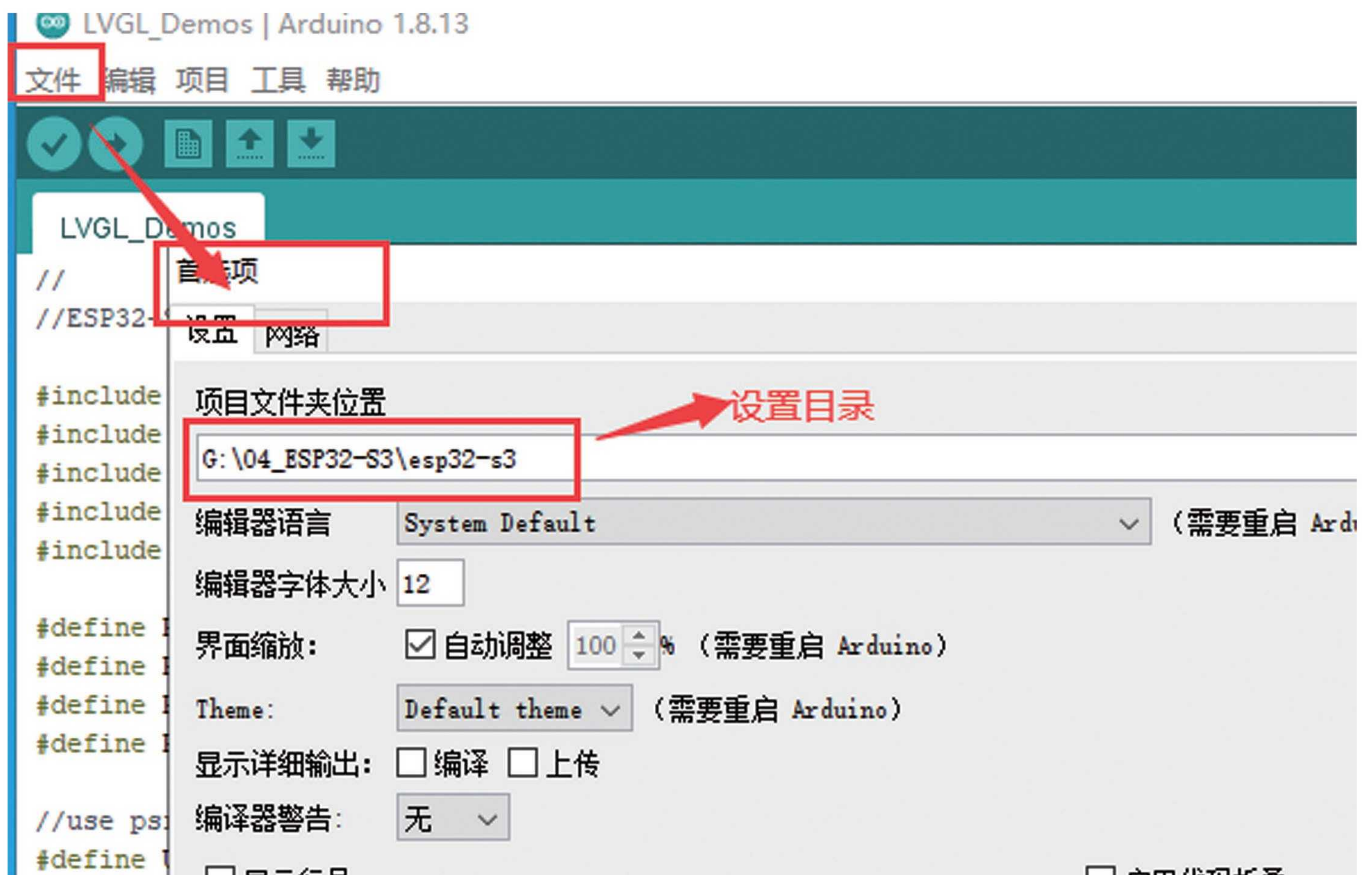
copy到libraries目录下

窗 > gy (G:) > 04_ESP32-S3 > esp32-s3 > libraries >

名称	修改日期	类型	大小
DHT11	2024/10/23 星期三 3:...	文件夹	
ESP32Servo	2024/10/23 星期三 3:...	文件夹	
ESP8266_and_ESP32_OLED_driver_for...	2024/10/23 星期三 3:...	文件夹	
Freenove_WS2812_Lib_for_ESP32	2024/10/23 星期三 3:...	文件夹	
lvgl	2024/10/23 星期三 3:...	文件夹	
TFT_eSPI	2024/10/23 星期三 3:...	文件夹	
TFT_Touch	2024/10/23 星期三 3:...	文件夹	
U8g2	2024/10/23 星期三 3:...	文件夹	

arduino设置目录

库

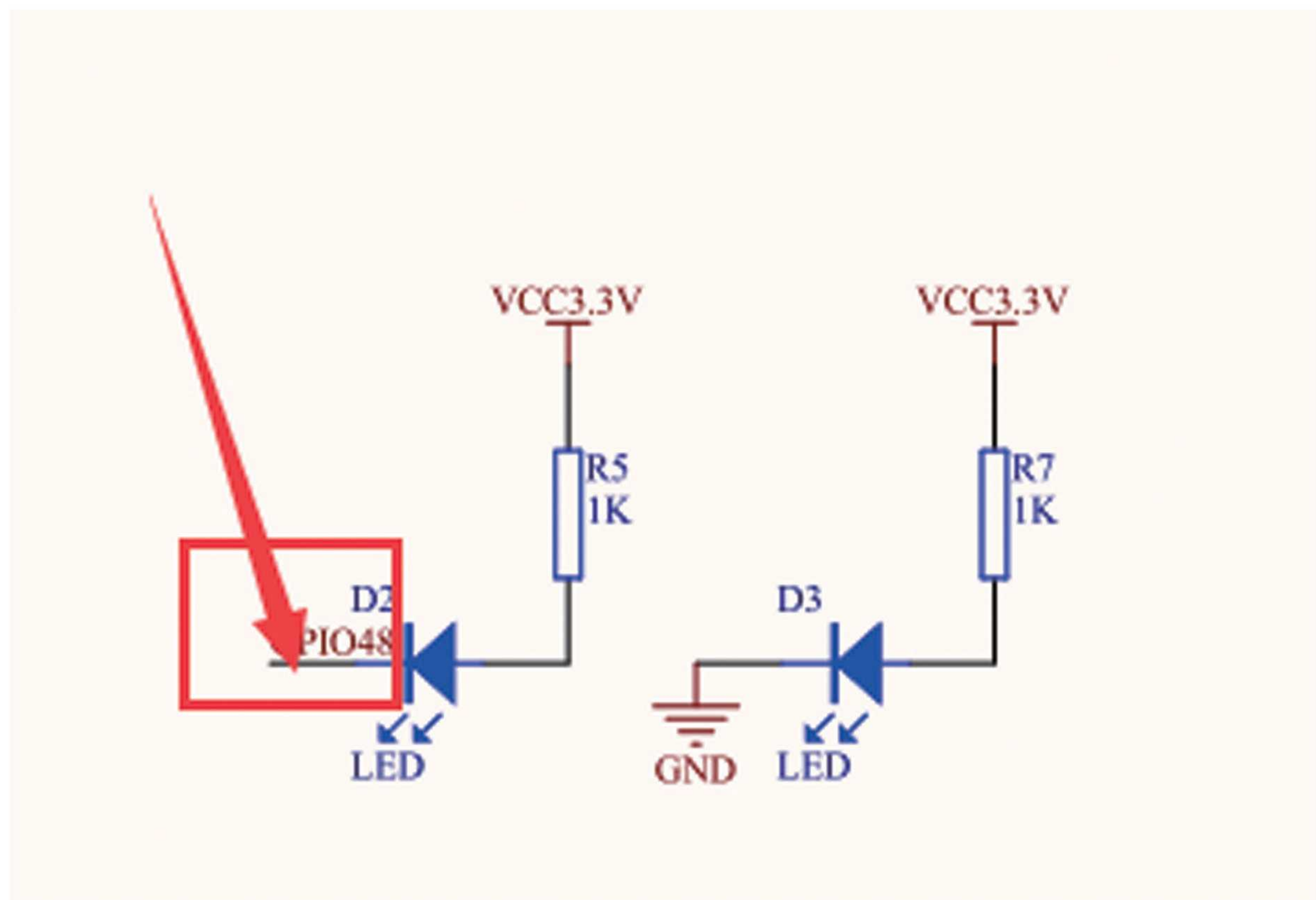


b) Routine Explanation (VSCode+Arduino)

If there is no expansion board, everyone can wire it themselves and connect it to external devices.

*LED control

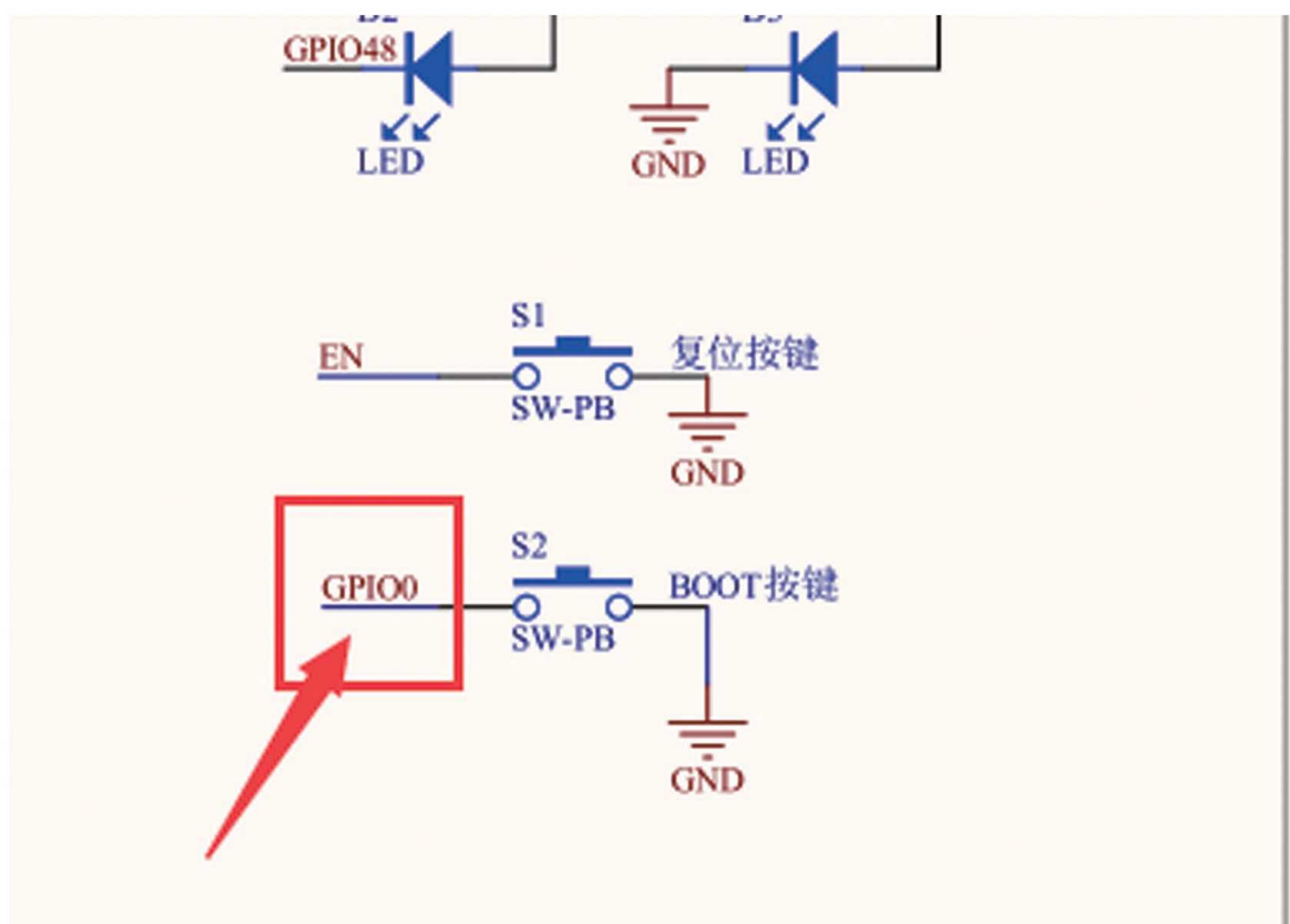
此电脑 > gy (G:) > ESP32-S3教程 > 核心板 > ESP32-S3-核心板 > 04-arduino教程和资料 > 03-arduino例程和库 > Arduino_Demo				
名称	修改日期	类型	大小	
1_LED	2024/10/27 星期日 2...	文件夹		
2_BUTTON	2024/10/27 星期日 2...	文件夹		
3_EXIT	2024/10/27 星期日 2...	文件夹		
4_Uart	2024/10/27 星期日 2...	文件夹		
5_PWM_LED	2024/10/27 星期日 2...	文件夹		



*Button routine

此电脑 > gy (G:) > ESP32-S3教程 > 核心板 > ESP32-S3-核心板 > 04-arduino教程和资料 > 03-arduino例程和库 > Arduino_Demo

名称	修改日期	类型	大小
1_LED	2024/10/27 星期日 2...	文件夹	
2_BUTTON	2024/10/27 星期日 2...	文件夹	
3_EXIT	2024/10/27 星期日 2...	文件夹	
4_Uart	2024/10/27 星期日 2...	文件夹	



Can't interrupt control problem troubleshooting?

Arduino routines use wiki link address: <https://docs.espressif.com/projects/arduino-esp32/en/latest/libraries.html>

- 1) The interrupt function requires the addition of the ARDUINO-ISR-ATTR macro
- 2) Debugging using serial port
- 3) Due to incorrect flash size settings, the program ran away during runtime

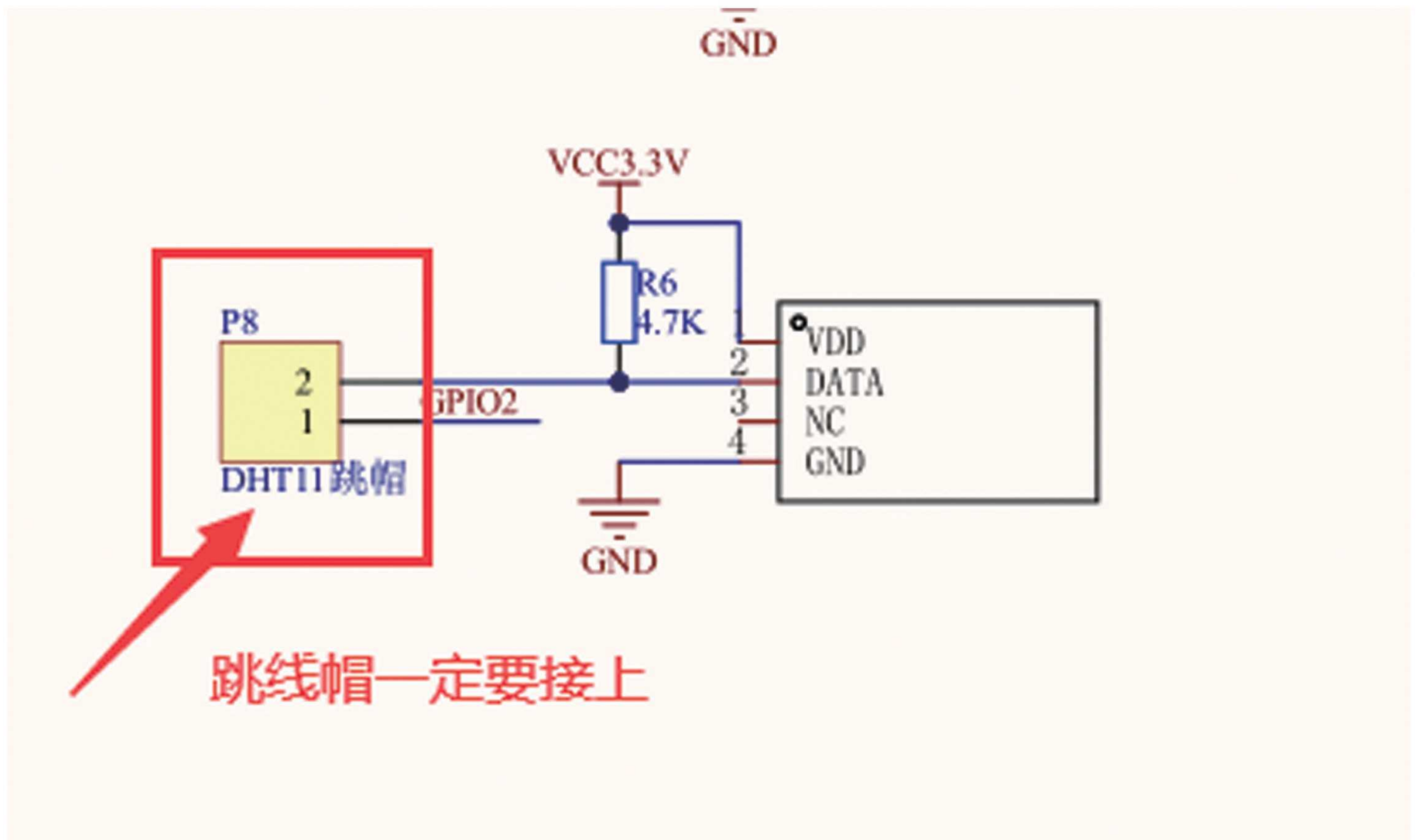
*Uart routine

名称	修改日期	类型	大小
1_LED	2024/10/27 星期日 2...	文件夹	
2_BUTTON	2024/10/27 星期日 2...	文件夹	
3_EXIT	2024/10/27 星期日 2...	文件夹	
4_Uart	2024/10/27 星期日 2...	文件夹	
5_PWM_LED	2024/10/27 星期日 2...	文件夹	
6_Servo	2024/10/27 星期日 2...	文件夹	
7_ADC	2024/10/27 星期日 2...	文件夹	
8_DHT11	2024/10/27 星期日 2...	文件夹	
9_IIC_OLED	2024/10/27 星期日 2...	文件夹	
10_TFT_SPI	2024/10/27 星期日 2...	文件夹	
11_LVGL	2024/10/27 星期日 2...	文件夹	
12_Cammera	2024/10/27 星期日 2...	文件夹	

$$2^{12}=4096$$

GPO1 cannot be reused with other functional pins

*DHT11 Temperature and Humidity Sensor Routine



DHT11 output data format:

8-bit humidity integer data+8-bit humidity decimal data+8-bit temperature integer data+8-bit temperature decimal part+8-bit checksum

byte4	byte3	byte2	byte1	byte0
00101101	00000000	00011100	00000000	01001001
整数	小数	整数	小数	校验和
湿度		温度		校验和

The values of humidity and temperature can be obtained from the above data, and the calculation method is:

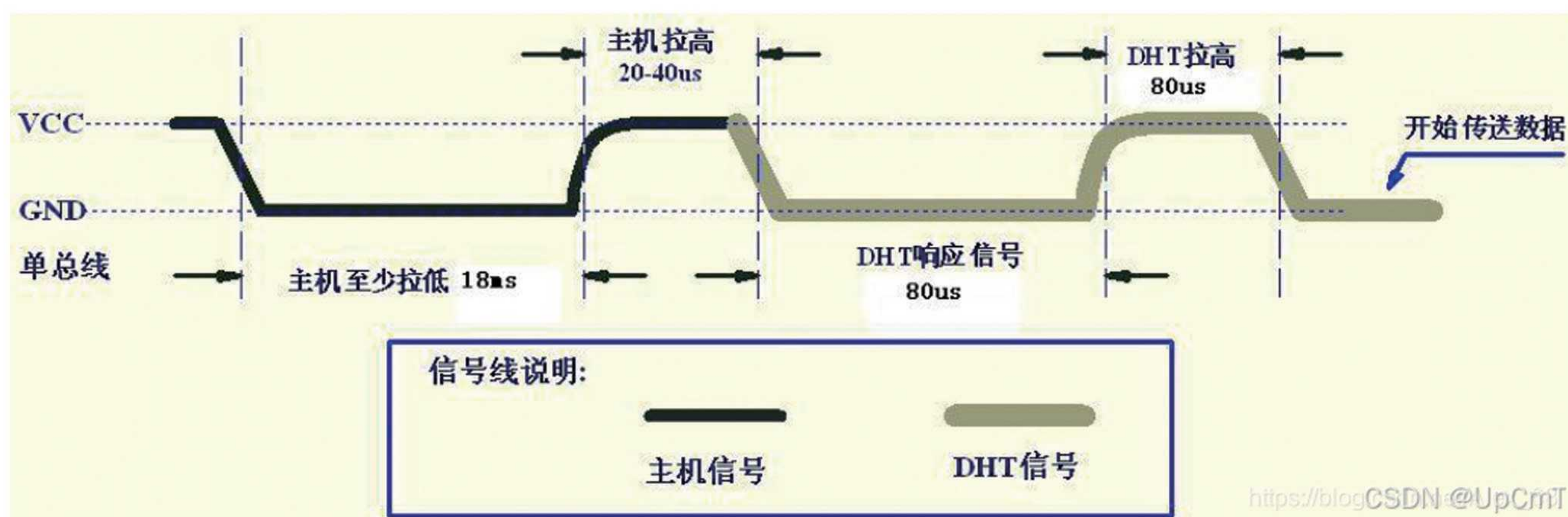
Humidity=byte4 byte3 = 45.0(%RH)

Temperature=byte2 byte1 = 28.0(°C)

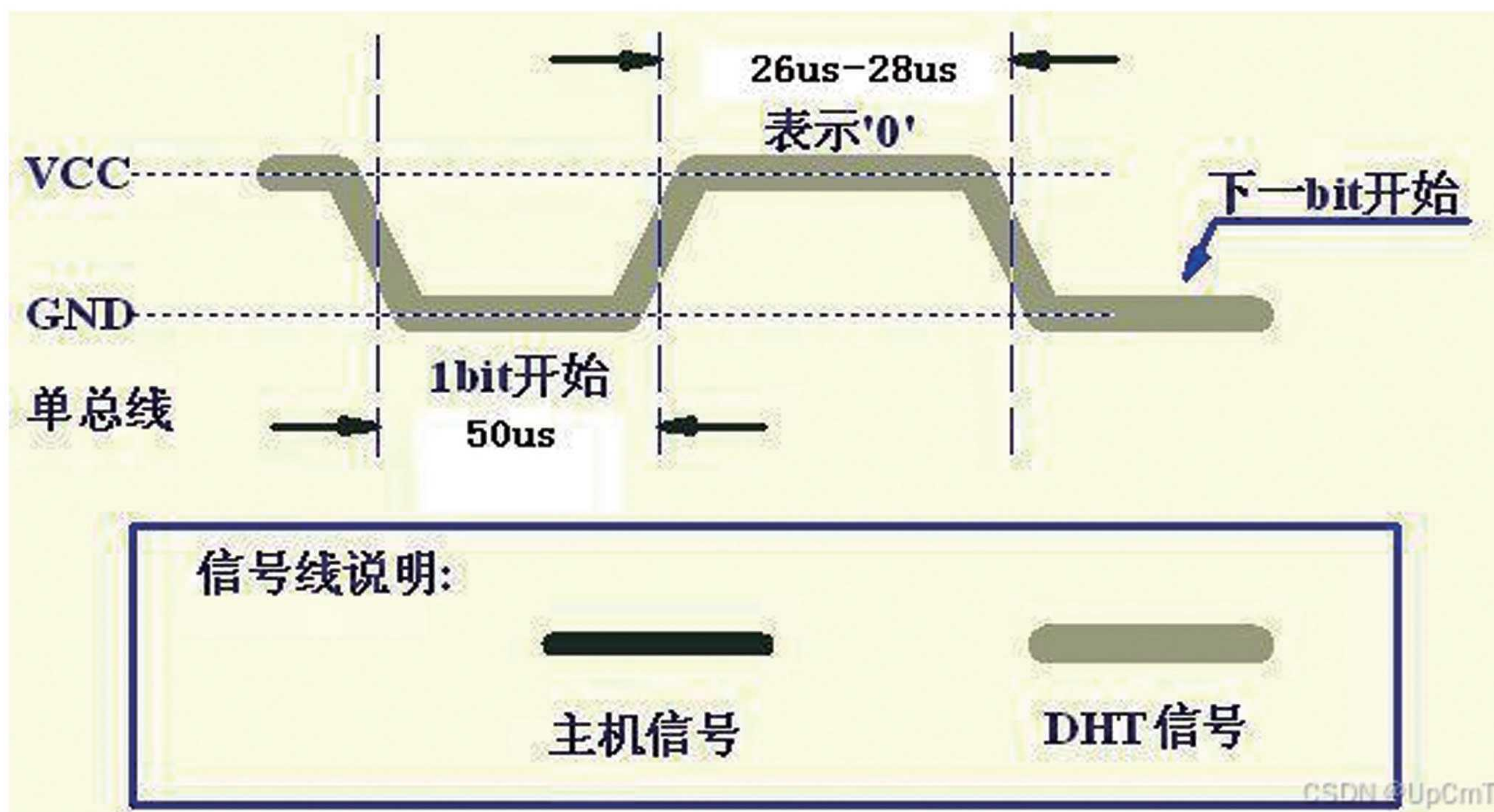
Verification (byte5)=? Byte4+byte3+byte2+byte1=73 (=humidity+temperature) (if they are equal, the verification is correct)

Sequence introduction: The host (ESP32) sends a data transmission command to DHT11 (pull low and then pull high) -> DHT11 responds to the received command (pull low and then pull high pin) -> DHT11 transmits data to the host (ESP32) (pull low and then pull high)

Command sending and response:



Data 0:



*OLED routine

1_LED	2024/10/27 星期日 2...	文件夹
2_BUTTON	2024/10/27 星期日 2...	文件夹
3_EXIT	2024/10/27 星期日 2...	文件夹
4_Uart	2024/11/1 星期五 2:10	文件夹
5_PWM_LED	2024/10/27 星期日 2...	文件夹
6_Servo	2024/10/27 星期日 2...	文件夹
7_ADC	2024/11/3 星期日 22:...	文件夹
8_DHT11	2024/11/3 星期日 23:...	文件夹
9_IIC_OLED	2024/11/1 星期五 2:42	文件夹
10_TFT_SPI	2024/10/27 星期日 2...	文件夹
11_LVGL	2024/10/27 星期日 2...	文件夹
12_Camera	2024/10/27 星期日 2...	文件夹



The pins of GPO41 and GPO42 cannot be occupied

Oled Screen Introduction:

1) Screen initialization

2) Position the cursor on the video memory

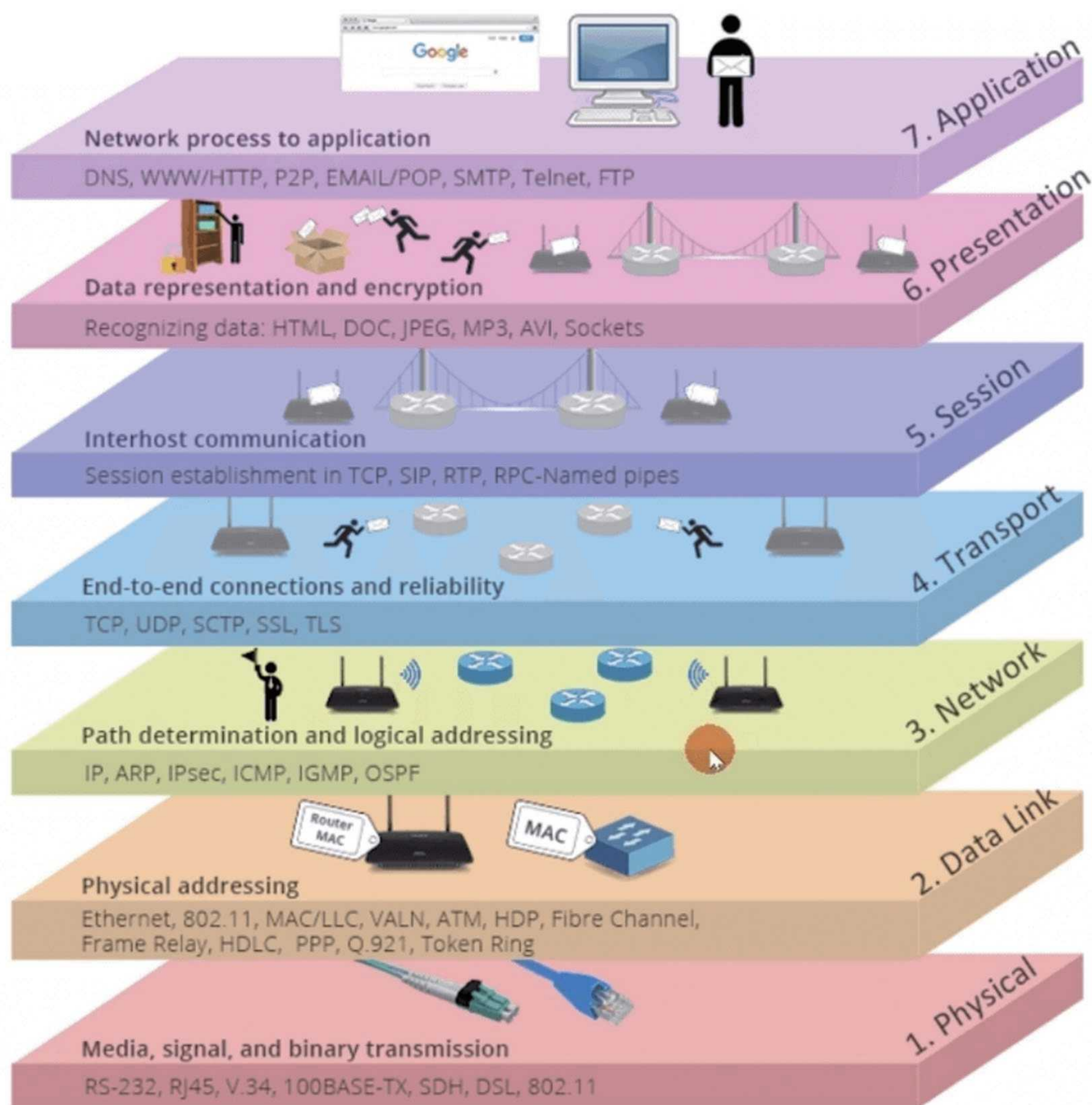
3) Write data to the cursor position, and after writing the data, the cursor will automatically move backward by 1 unit

Here is the layout diagram of the video memory: 64X128 bits

64			0 列	1 列	2 列	3 列		124 列	125 列	126 列	127 列
Page0	bit0	0行	0								
	bit1	1行	0								
	bit2	2行	0								
	bit3	3行	1								
	bit4	4行	0								
	bit5	5行	0								
	bit6	6行	0								
	bit7	7行	0								
Page1		8到 15行									
Page2		16到 23行									
Page3											
Page4											
Page5											
Page6											
Page7		56行 到63 行									

CSDN @单行梦想家

- 3) The data written to the OLED screen is divided into two types,
Command type: such as controlling cursor position, 0x00+command value
Data type: such as write cached value, 0x01+data value
- 4) Display principle: Allocate a buffer of the same size as the screen [size: 64x128/8 bytes], modify the contents of the buffer, and directly send all the contents of the buffer to the video memory, refreshing all columns of the video memory
- Wifi routine
- Seven layer network model:



- 应用层

- 表示层

- 会话层

- 传输层

- 网络层

- 数据链路层

- 物理层

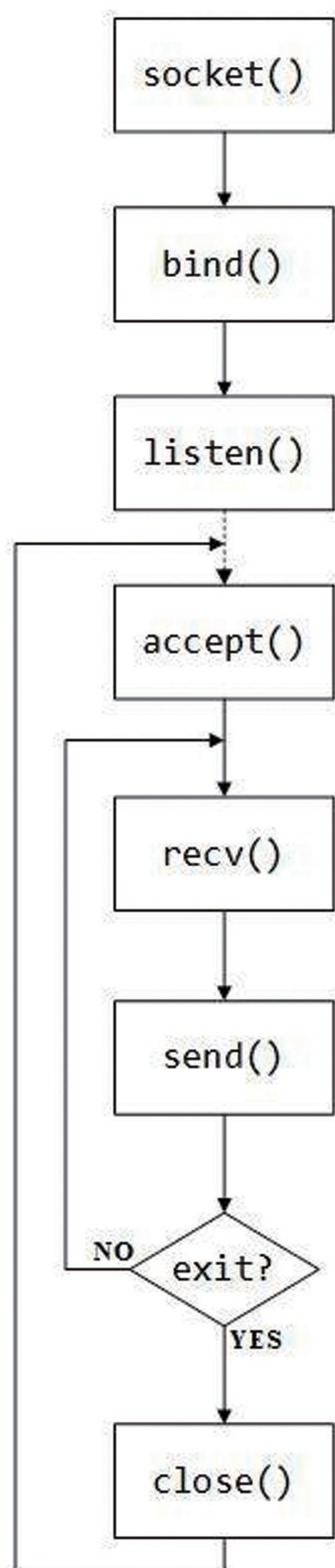
CSDN @RS迷途小书童

Memory method: "Internet of Things, Numbers, Communication, Meeting, and Correspondence". Remember these 7 words, and you will remember these 7 layers conform to

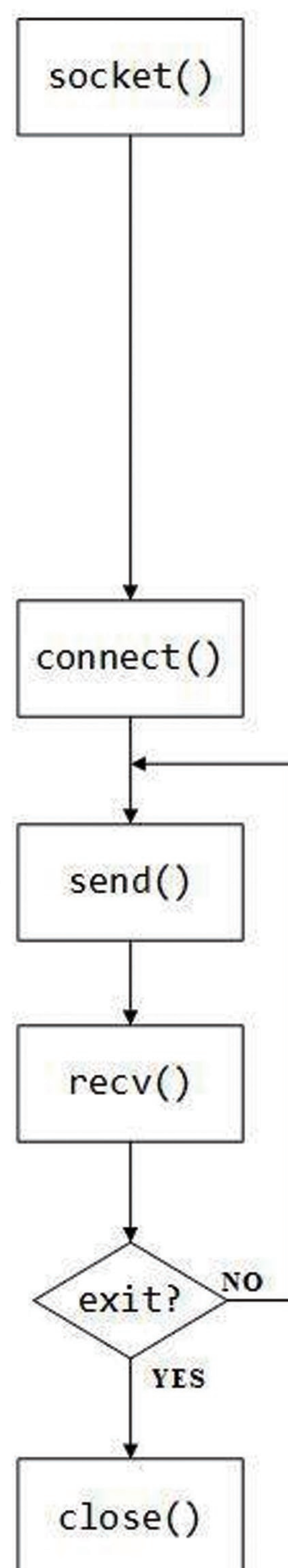
The data is ultimately transmitted to the hardware network through the data link layer and physical layer, and the data at both the data link layer and physical layer is encapsulated layer by layer.

1) TCP communication process

TCP 服务器



TCP 客户端

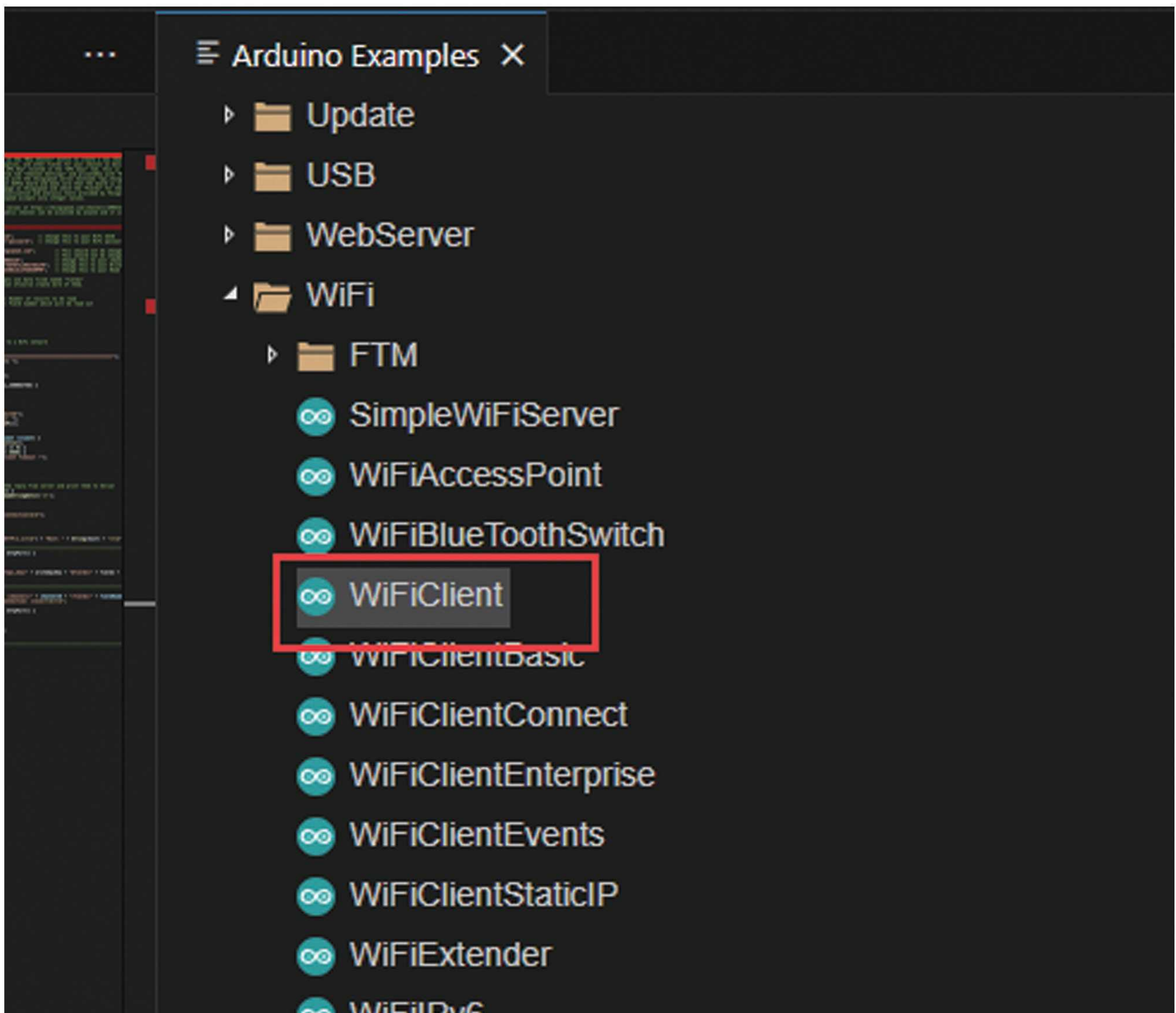


服务器阻塞至客户端请求连接

客户端向服务器发送数据

服务器向客户端响应数据

2) tcp client



HTTP and MQTT are protocols built on top of TCP.

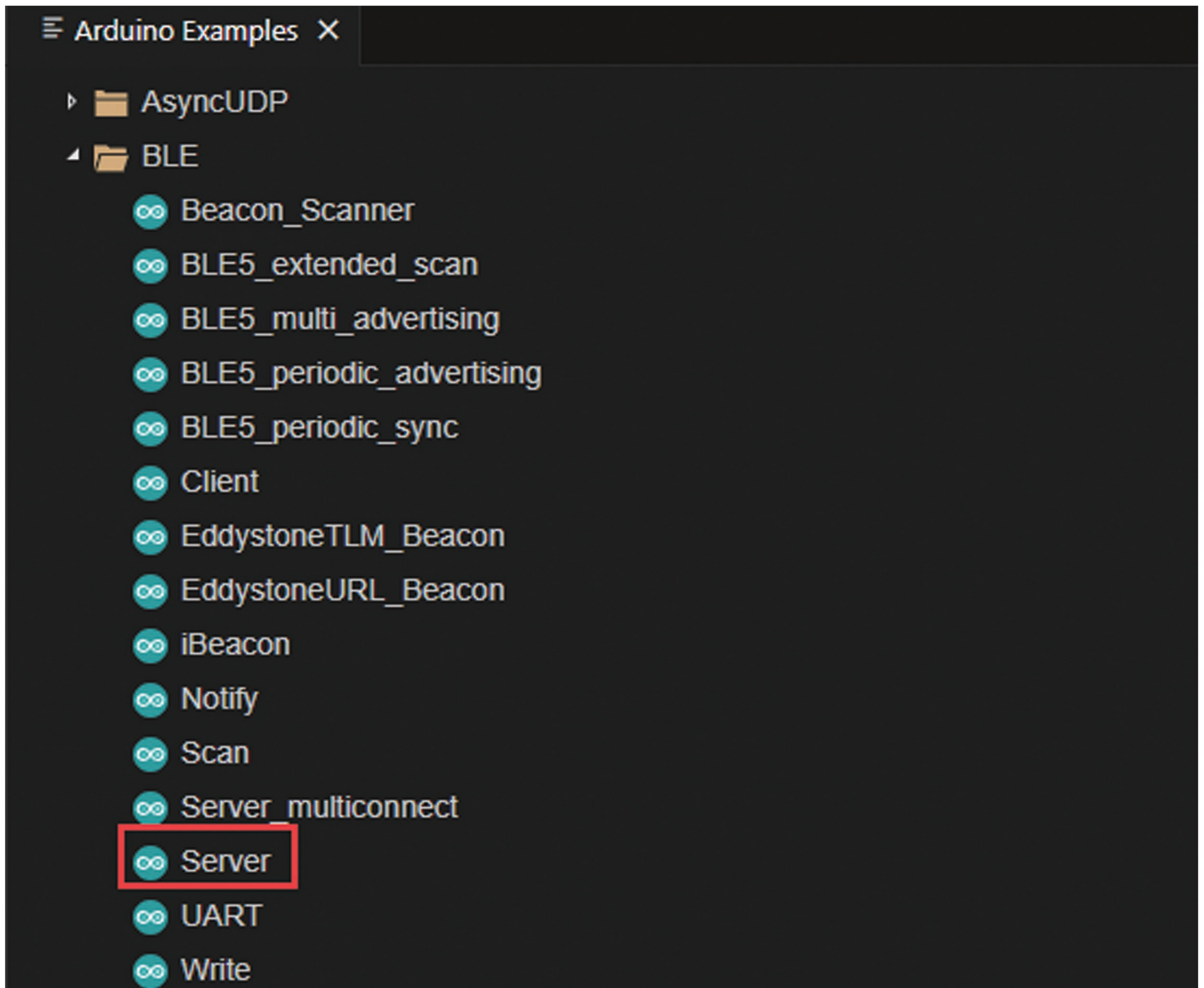
*BLE routine

Our previous tutorial, if you are interested, you can watch it. If you have any questions, you can communicate in the group.

Reuse of previous tutorials,

 01-arduino依赖库copy.mp4	2024/10/27 星期日 2...	MP4 文件	11,304 KB
 02-led例程讲解.mp4	2024/10/27 星期日 2...	MP4 文件	21,363 KB
 03-Button例程.mp4	2024/10/27 星期日 2...	MP4 文件	38,523 KB
 04-排查按键中断不生效问题.mp4	2024/11/1 星期五 2:07	MP4 文件	39,344 KB
 05-串口例程演示.mp4	2024/11/3 星期日 22:...	MP4 文件	75,627 KB
 06-ADC例程演示.mp4	2024/11/3 星期日 22:...	MP4 文件	33,243 KB
 07-DHT11数据读取介绍.mp4	2024/11/3 星期日 23:...	MP4 文件	28,285 KB
 08-DHT11例程分析.mp4	2024/11/3 星期日 23:...	MP4 文件	37,707 KB
 09-oled例程分析.mp4	2024/11/4 星期一 0:27	MP4 文件	68,381 KB
 10-网络通讯协议介绍.mp4	2024/11/4 星期一 1:06	MP4 文件	44,753 KB
 11-wificlient分析.mp4	2024/11/4 星期一 1:43	MP4 文件	59,976 KB
 12-蓝牙server相关的使用之一.wmv	2024/11/4 星期一 1:50	WMV 文件	206,218 KB
 13-通知怎么被订阅以及server实验.wmv	2024/11/4 星期一 1:50	WMV 文件	161,741 KB
 14-service的特征值的读写.wmv	2024/11/4 星期一 1:48	WMV 文件	52,790 KB
 15-蓝牙广播和连接理论知识.wmv	2024/11/4 星期一 1:50	WMV 文件	205,992 KB
 16-client编程思路.wmv	2024/11/4 星期一 1:48	WMV 文件	23,024 KB
 17-client编程实现代码分析.wmv	2024/11/4 星期一 1:50	WMV 文件	153,651 KB
 18-怎么获取server的service和characteristic.wmv	2024/11/4 星期一 1:50	WMV 文件	108,149 KB

复用之前例程大家感兴趣
可以研究一下



Apple phones can download the following software for debugging:



Android can directly install software:

电脑 > gy (G:) > ESP32-S3教程 > 核心板 > ESP32-S3-核心板 > 07_蓝牙调试助手



设备列表



搜索



Long name works now

1 Service Advertised

-35 dBm

846C1CA2-719B-E7B2-7E0B-28F7A47CC6B7



ZZYX_LOCK

1 Service Advertised

-93 dBm

ACE4810B-EF1B-BD1C-1919-24CD40CB9AC5



ZZYX_LOCK

1 Service Advertised

-93 dBm

B8FD515C-22F3-1105-FA5A-B147A57907F4



ZZYX_LOCK

1 Service Advertised

-94 dBm

EFBC03F2-9941-C040-76AF-CCFCD4F7579C



ZZYX_LOCK

1 Service Advertised

-89 dBm

899FE64F-D886-C457-E102-F25B6DA72495



未命名设备

-



设备



日志

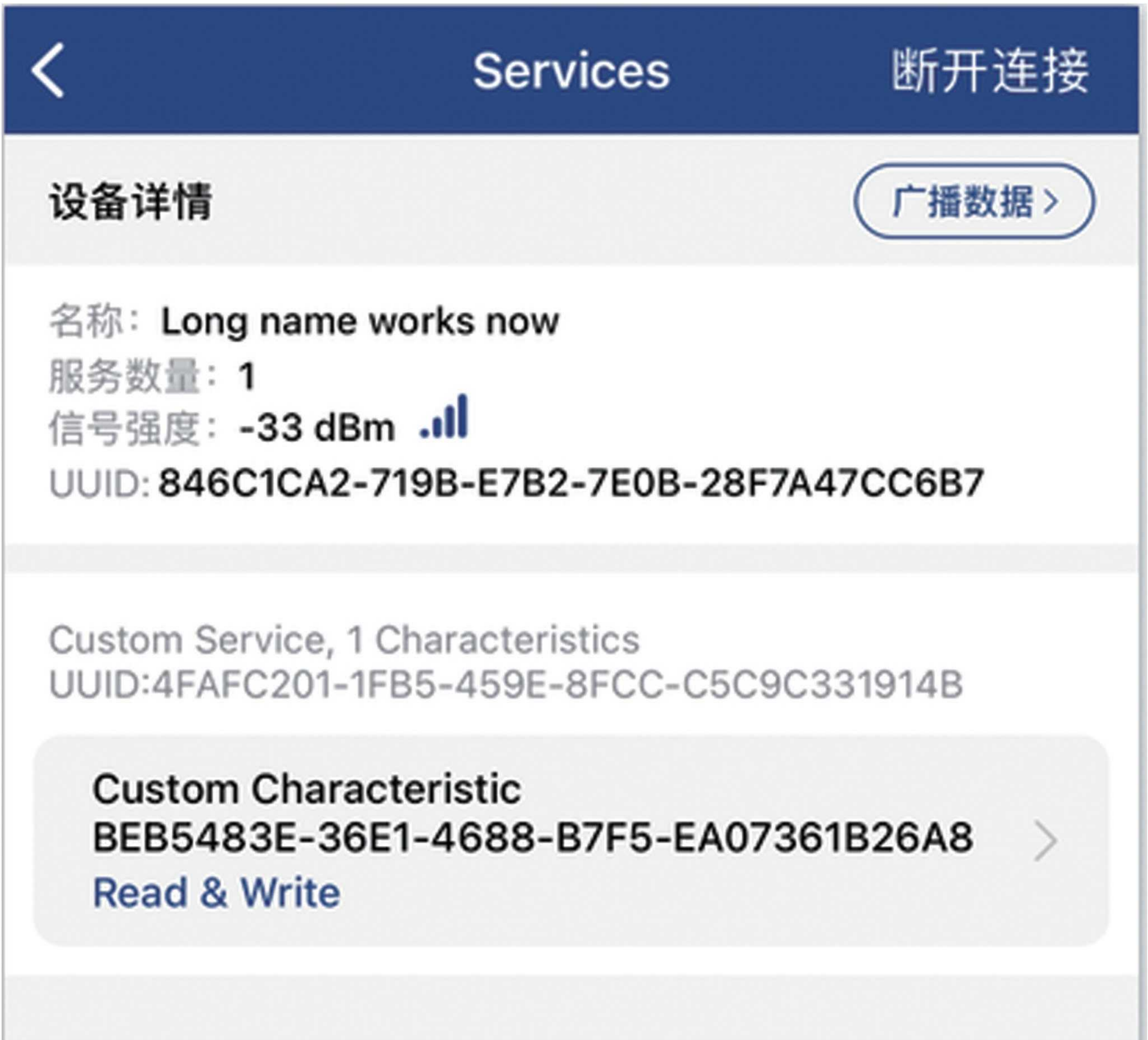


数据包



设置

After clicking on the Bluetooth name, it will automatically connect to Bluetooth:



Click on 'Custom Characteristic' and select 'read' to read the user-defined values,

< Services

Characteristic Detail

日志

Service: Custom Service

4FAFC201-1FB5-459E-8FCC-C5C9C331914B

Characteristic: Custom Characteristic

BEB5483E-36E1-4688-B7F5-EA07361B26A8

写数据:

Hex

ASCII

填写需要发送的数据 (ASCII)

Write

选择批量数据

批量写入

读数据:

Hex

ASCII

Hello World says Neil

×

Notify

Read

After writing it, reading it out again is the value written by the user,



*About TFT LCD Video Tutorial Instructions

The Lcd routine will be produced in conjunction with the vgl tutorial!!

2、 Micropathon Video Tutorial

3、 Idf Video Tutorial